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MIL-M-28787/512
15 SEPTEMBER 1989

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MEMORY MAPPED I/O MODULE, KEY CODE BWG

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic Memory Mapped Input/Output (MMIO) module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

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3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 2.751 failures per million operating hours for class I or 6.890 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be BWG.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	Class I and II 25 ± 5°C 4/		Class I 0 ± 5°C 4/ 60 ± 5°C		Class II -55 ± 5°C 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
B1-B6	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Vcc stress	IV	S	---------Module shall pass all remaining tests---------						
D1	Ibd	V	S	-0.12	0.22	-0.12	0.22	-0.12	0.22	mA
D2-D9	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D10	Ibd	V	S	-0.12	0.32	-0.12	0.32	-0.12	0.32	mA
D11-D13	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D14,D15	Ibd	V	S	0.0	0.3	0.0	0.3	0.0	0.3	mA
D16	Ibd	V	S	-0.12	0.22	-0.12	0.22	-0.12	0.22	mA
D17	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D18	Ibd	V	S	0.0	0.3	0.0	0.3	0.0	0.3	mA
D19-D21	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D22	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	mA
D23-D30	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D31	Ibd	V	S	-0.12	0.72	-0.12	0.72	-0.12	0.72	mA
D32-D35	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D36	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D37	Ibd	V	S	0.0	0.3	0.0	0.3	0.0	0.3	mA
D38-D50	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D51	Ibd	V	S	-0.12	0.42	-0.12	0.42	-0.12	0.42	mA
E1-E37	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E38	Ios	VI	S	-225.0	-100.0	-225.0	-100.0	-225.0	-100.0	mA
F1	Iih	VII	X	-6.03	-2.85	-6.03	-2.85	-6.03	-2.85	mA
F2-F9	Iih	VII	X	-0.36	0.02	-0.36	0.02	-0.36	0.02	mA
F10	Iih	VII	X	-6.63	-2.83	-6.63	-2.83	-6.63	-2.83	mA
F11-F13	Iih	VII	X	-0.25	0.07	-0.25	0.07	-0.25	0.07	mA
F14-F18	Iih	VII	X	-0.61	0.09	-0.61	0.09	-0.61	0.09	mA
F19-F21	Iih	VII	X	-0.36	0.02	-0.36	0.02	-0.36	0.02	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	Class I and II 25 ± 5°C 4/		Class I 0 ± 5°C 4/ 60 ± 5°C		Class II -55 ± 5°C 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
F22,F23	Iih	VII	X	-1.8	0.06	-1.8	0.06	-1.8	0.06	mA
F24	Iih	VII	X	-5.63	-2.85	-5.63	-2.85	-5.63	-2.85	mA
F25-F30	Iih	VII	X	-0.61	0.09	-0.61	0.09	-0.61	0.09	mA
F31,F32	Iih	VII	X	-0.85	0.09	-0.85	0.09	-0.85	0.09	mA
F33	Iih	VII	X	-0.4	0.04	-0.4	0.04	-0.4	0.04	mA
F34	Iih	VII	X	-1.8	0.06	-1.8	0.06	-1.8	0.06	mA
F35,F36	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F37	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F38	Iih	VII	X	-4.0	0.1	-4.0	0.1	-4.0	0.1	mA
F39-F46	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F47	Iih	VII	X	-6.63	-2.75	-6.63	-2.75	-6.63	-2.75	mA
F48-F51	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F52	Iih	VII	X	-0.6	0.04	-0.6	0.04	-0.6	0.04	mA
F53	Iih	VII	X	-1.0	0.06	-1.0	0.06	-1.0	0.06	mA
F54-F66	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F67	Iih	VII	X	-1.05	0.11	-1.05	0.11	-1.05	0.11	mA
F68	Iih	VII	X	-6.03	-2.81	-6.03	-2.81	-6.03	-2.81	mA
G1	Iil	VIII	X	-6.03	-2.85	-6.03	-2.85	-6.03	-2.85	mA
G2-G9	Iil	VIII	X	-0.36	0.02	-0.36	0.02	-0.36	0.02	mA
G10	Iil	VIII	X	-6.63	-2.83	-6.63	-2.83	-6.63	-2.83	mA
G11-G13	Iil	VIII	X	-0.25	0.07	-0.25	0.07	-0.25	0.07	mA
G14-G18	Iil	VIII	X	-0.61	0.09	-0.61	0.09	-0.61	0.09	mA
G19-G21	Iil	VIII	X	-0.36	0.02	-0.36	0.02	-0.36	0.02	mA
G22,G23	Iil	VIII	X	-1.8	0.06	-1.8	0.06	-1.8	0.06	mA
G24	Iil	VIII	X	-5.63	-2.85	-5.63	-2.85	-5.63	-2.85	mA
G25-G30	Iil	VIII	X	-0.61	0.09	-0.61	0.09	-0.61	0.09	mA
G31,G32	Iil	VIII	X	-0.85	0.09	-0.85	0.09	-0.85	0.09	mA
G33	Iil	VIII	X	-0.4	0.04	-0.4	0.04	-0.4	0.04	mA
G34	Iil	VIII	X	-1.8	0.06	-1.8	0.06	-1.8	0.06	mA
G35,G36	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G37	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G38	Iil	VIII	X	-4.0	0.1	-4.0	0.1	-4.0	0.1	mA
G39-G46	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G47	Iil	VIII	X	-6.63	-2.75	-6.63	-2.75	-6.63	-2.75	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	Class I and II 25 ± 5°C 4/		Class I 0 ± 5°C 4/ 60 ± 5°C		Class II -55 ± 5°C 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
G48-G51	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G52	Iil	VIII	X	-0.6	0.04	-0.6	0.04	-0.6	0.04	mA
G53	Iil	VIII	X	-1.0	0.06	-1.0	0.06	-1.0	0.06	mA
G54-G66	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G67	Iil	VIII	X	-1.05	0.11	-1.05	0.11	-1.05	0.11	mA
G68	Iil	VIII	X	-6.03	-2.81	-6.03	-2.81	-6.03	-2.81	mA
H1-H38	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J42	Vol	X	X	0.0	0.5	0.0	0.5	0.0	0.5	V
K1,K2	Iozh	XI	X	0.0	0.25	0.0	0.25	0.0	0.25	mA
K3-K16	Iozh	XI	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA
K17	Iozh	XI	X	-0.1	0.1	-0.1	0.1	-0.1	0.1	mA
K18-K31	Iozl	XI	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA
K32	Iozl	XI	X	-0.1	0.1	-0.1	0.1	-0.1	0.1	mA
L1-L11	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L12,L13	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L14	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L15-L17	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L18	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L19	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L20-L27	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L28-L31	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L32-L46	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
M1	PSFC	XIII	S	11.0	20.6	11.0	20.6	11.0	20.6	uF
M2	PSFC	XIII	S	6.6	12.4	6.6	12.4	6.6	12.4	uF
N	Icc	XIV	X	750	1800	750	1800	750	1800	mA
P1,P2	Functional	XV	X	See table XV						

- See footnotes at end of table.

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TABLE I. *Module electrical requirements* - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	Class I and II 25 ± 5°C 4/		Class I 0 ± 5°C 4/ 60 ± 5°C		Class II -55 ± 5°C 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
Q1,Q2	Dynamic	XVI	S	----------See Table XVI----------						
Q3,Q4	Dynamic	XVI	Q	----------See Table XVI----------						
R1	Ci/o	XVIII	Q	4	100	4	100	4	100	pF
R2-R9	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R10	Ci/o	XVIII	Q	6	100	6	100	6	100	pF
R11	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R12	Ci/o	XVIII	Q	2	50	2	50	2	50	pF
R13	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R14,R15	Ci/o	XVIII	Q	6	100	6	100	6	100	pF
R16	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
R17	Ci/o	XVIII	Q	4	125	4	125	4	125	pF
R18	Ci/o	XVIII	Q	6	100	6	100	6	100	pF
R19,R20	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R21	Ci/o	XVIII	Q	2	100	2	100	2	100	pF
R22	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
R23-R30	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R31	Ci/o	XVIII	Q	4	200	4	200	4	200	pF
R32-R35	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R36	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
R37	Ci/o	XVIII	Q	6	75	6	75	6	75	pF
R38	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R39	Ci/o	XVIII	Q	2	50	2	50	2	50	pF
R40-R49	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R50	Ci/o	XVIII	Q	2	100	2	100	2	100	pF
R51	Ci/o	XVIII	Q	6	175	6	175	6	175	pF
R52-R54	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
R55-R57	Ci/o	XVIII	Q	6	75	6	75	6	75	pF
R58-R63	Ci/o	XVIII	Q	6	100	6	100	6	100	pF
R64	Ci/o	XVIII	Q	6	125	6	125	6	125	pF
R65	Ci/o	XVIII	Q	6	100	6	100	6	100	pF
R66,R67	Ci/o	XVIII	Q	6	125	6	125	6	125	pF
T1,T2	PSTA	XIX	Q	0	400	0	400	0	400	mV

- See footnotes at end of table.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/512-1 for class I and M28787/512-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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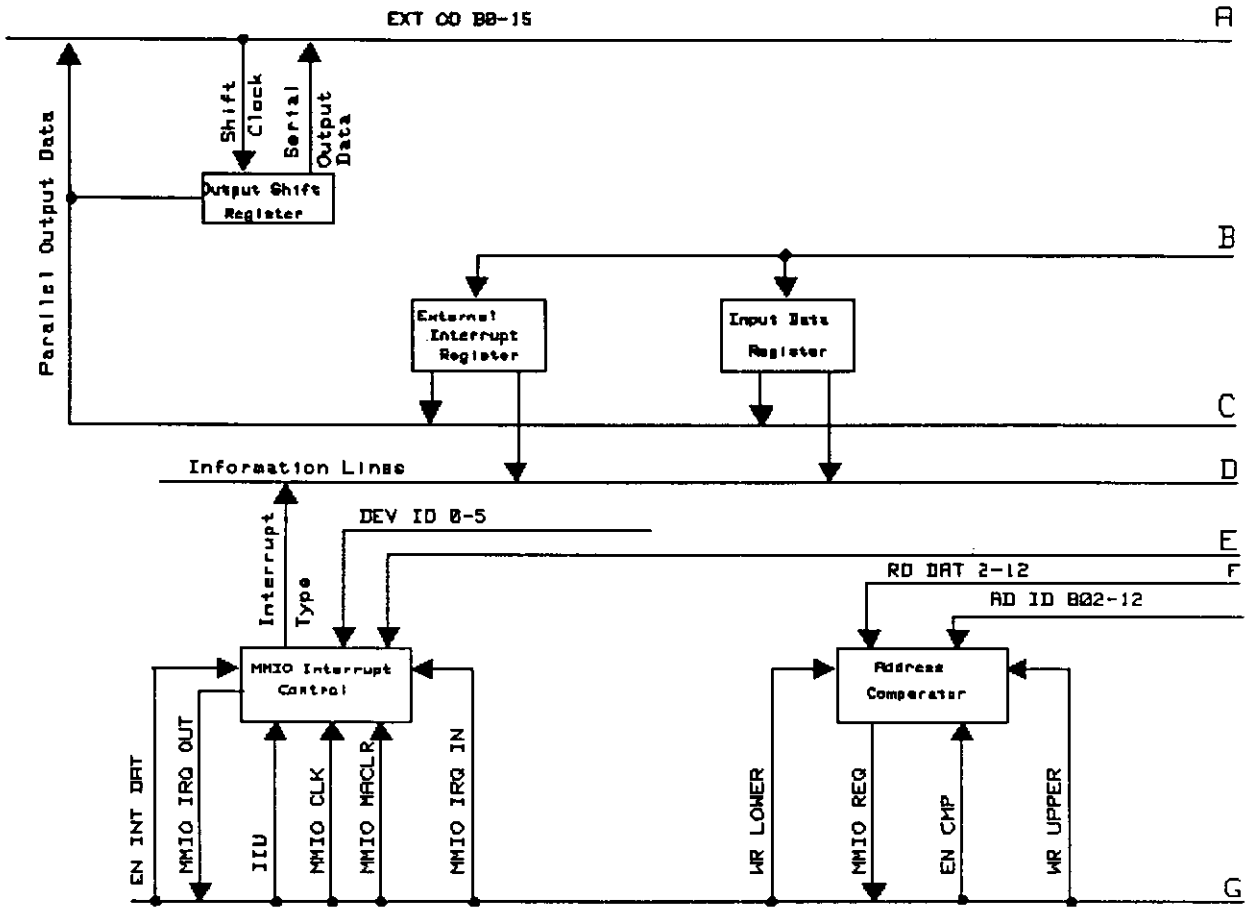


FIGURE 1. BWG functional block diagram.

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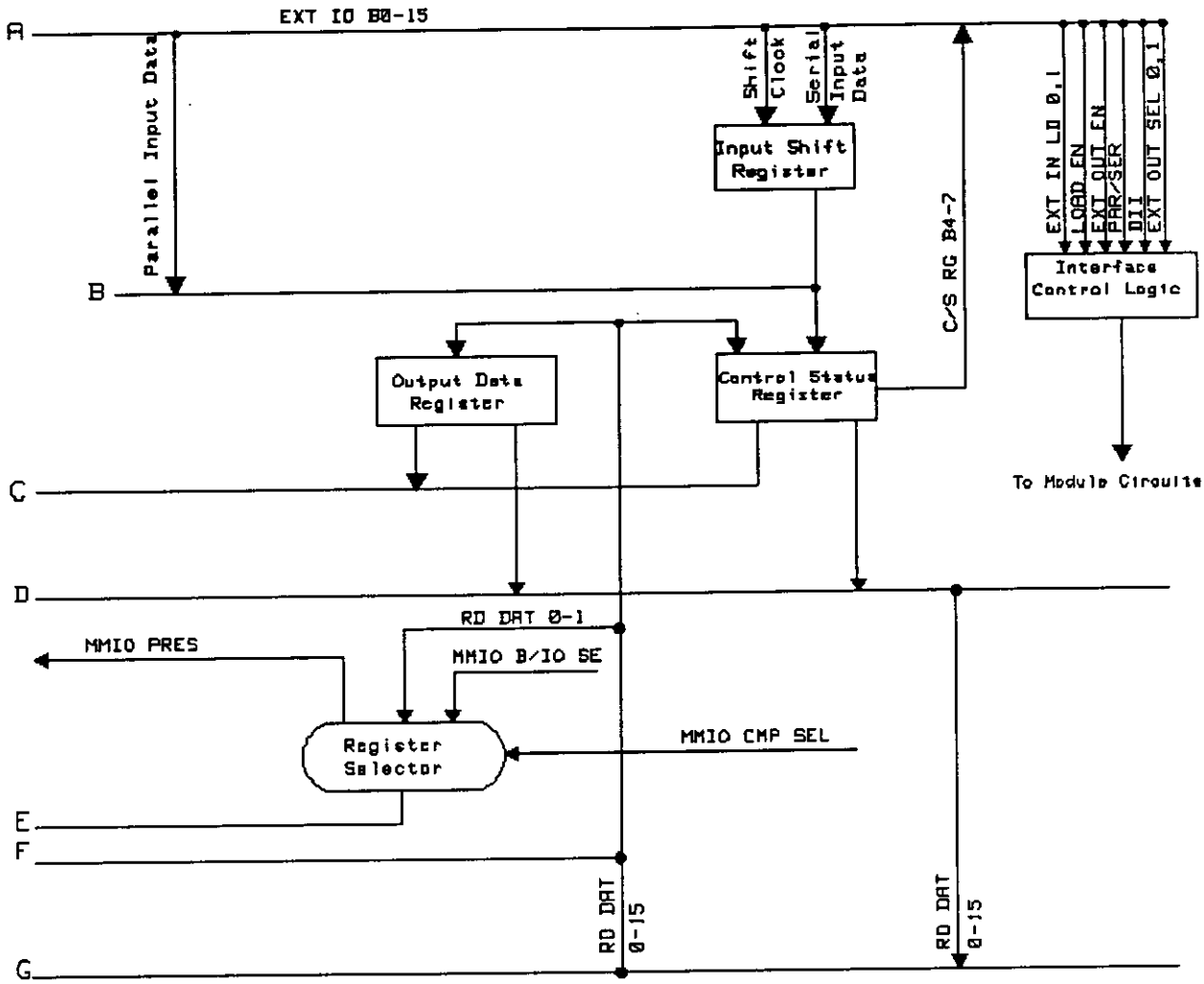


FIGURE 1. BWG functional block diagram - Continued.

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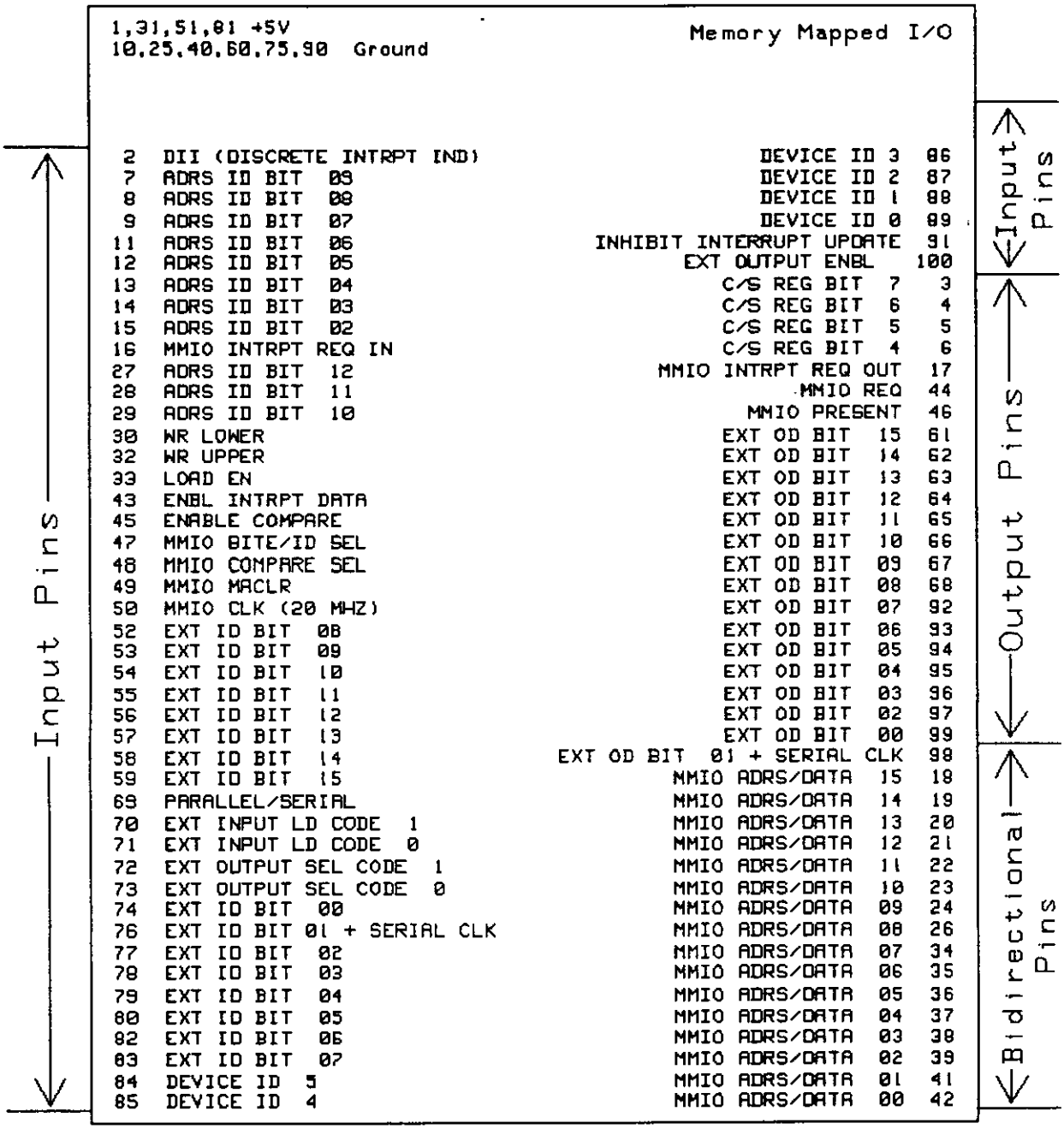


FIGURE 2. BWG pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, and L are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or L are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, and L where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

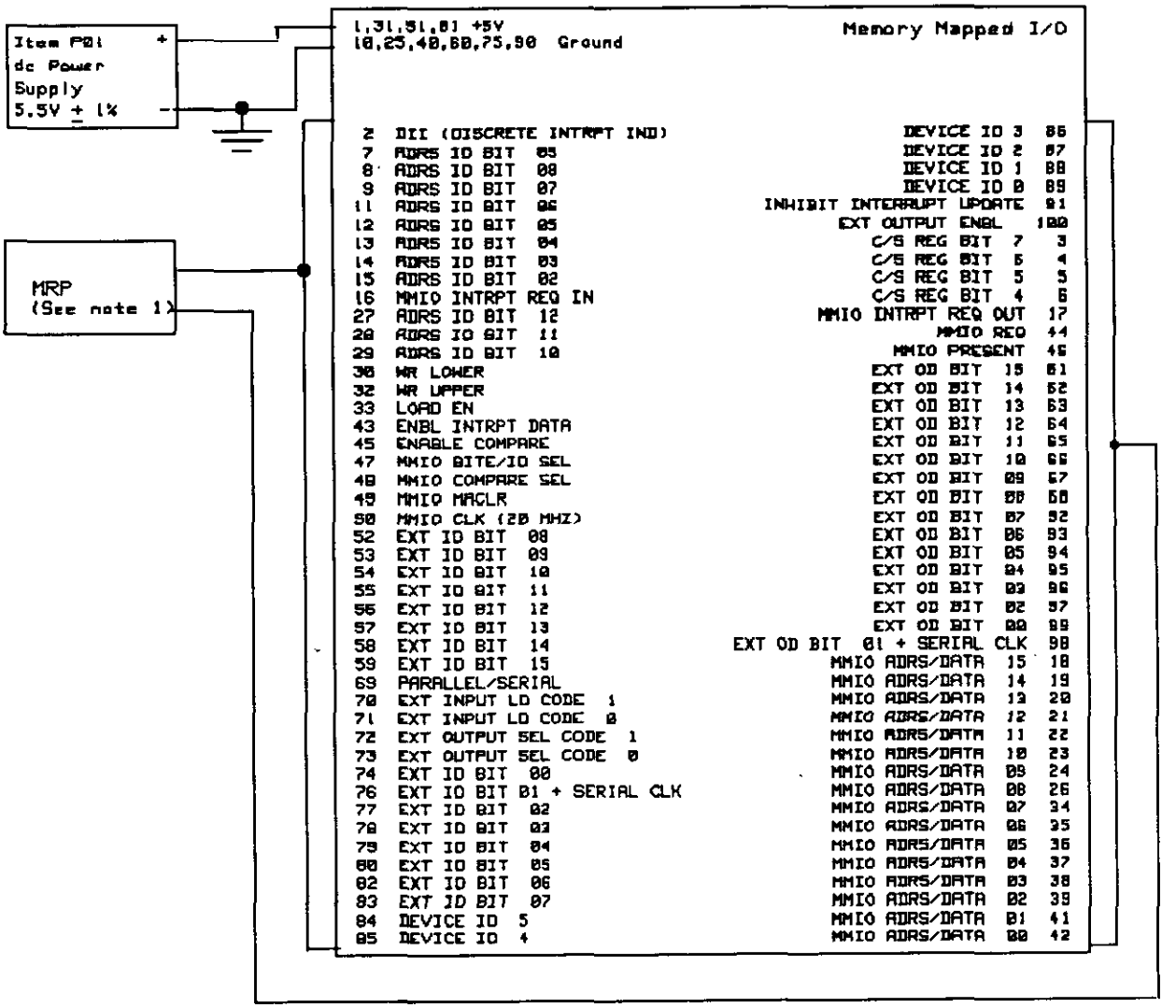
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

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NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. BWG operating temperature and component temperature test circuit.

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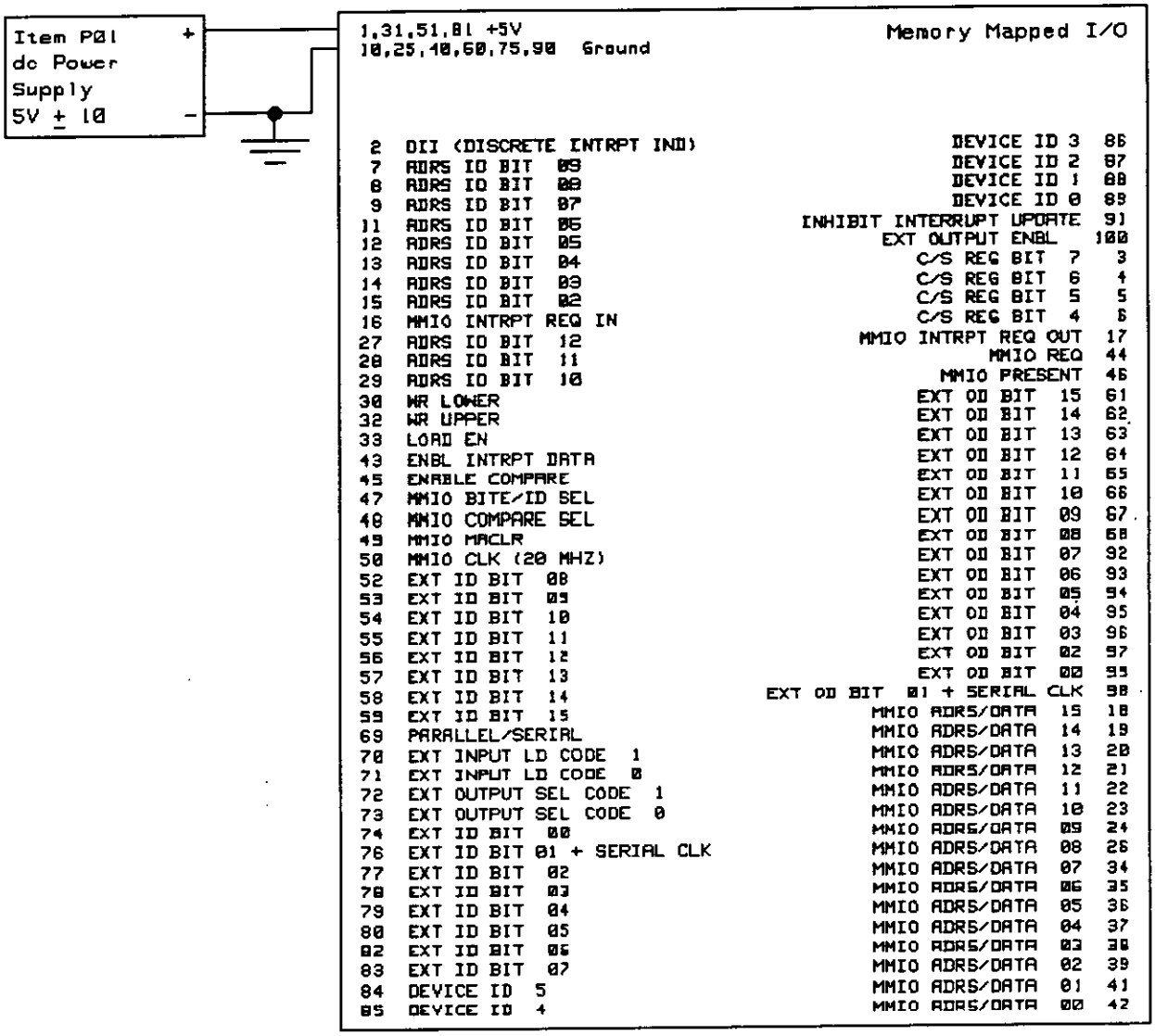


FIGURE 4. BWG vibration test circuit.

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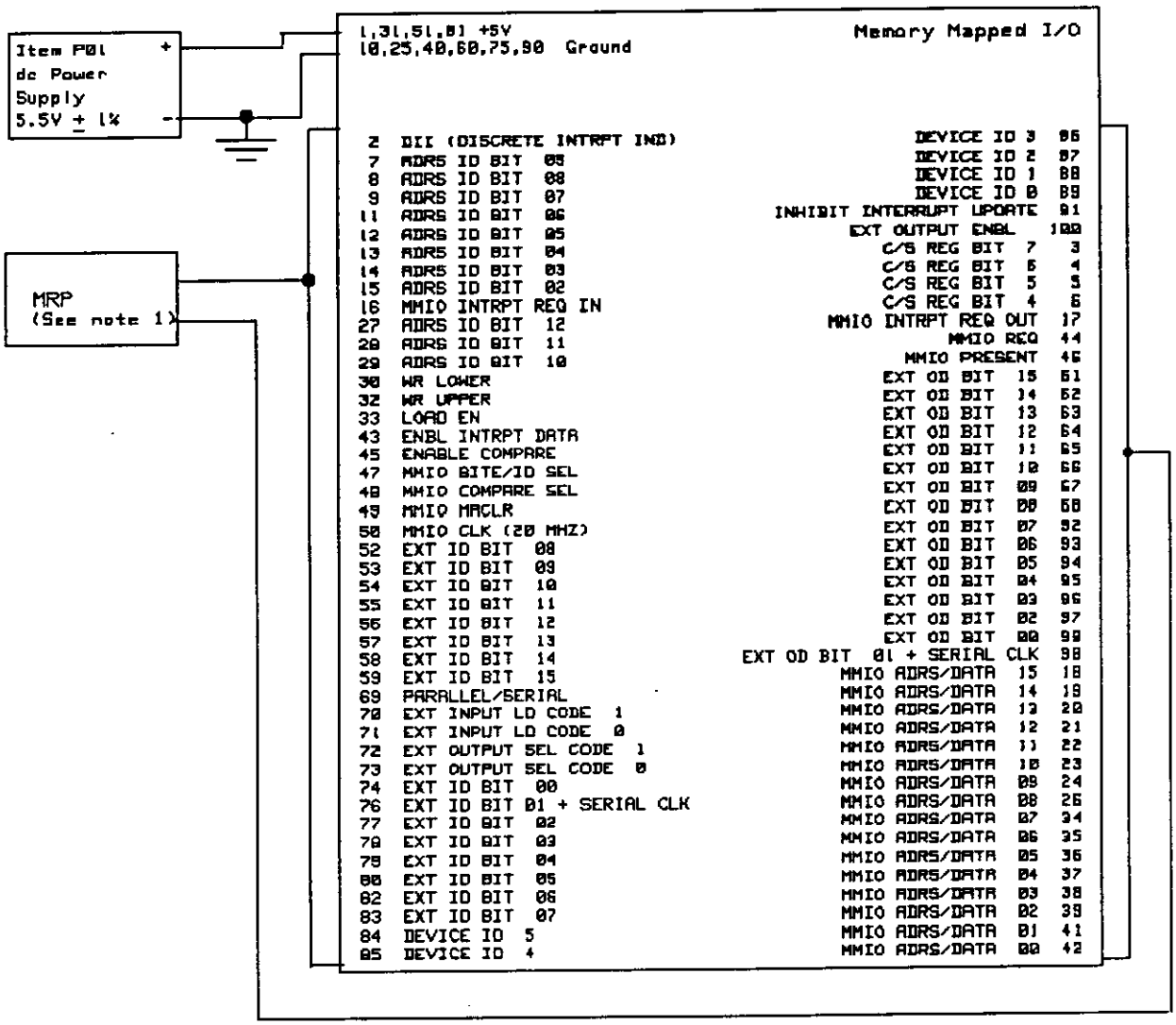
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. BWG life test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ci/o = input/output capacitance.
(c) Cout = output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Iih = input high level current.
(g) Iil = input low level current.
(h) Iin = input forcing current.
(i) Ioh = output high level current.
(j) Iol = output low level current.
(k) Ios = output short circuit current.
(l) Iozh = output high level leakage current.
(m) Iozl = output low level leakage current.
(n) NA = not applicable.
(o) PSFC = power supply filter capacitance.
(p) PSTA = power supply transient amplitude.
(q) Vic = input clamp diode voltage.
(r) Vih = input high level voltage.
(s) Vil = input low level voltage.
(t) Vin = input forcing voltage.
(u) Voh = output high level voltage.
(v) Vol = output low level voltage.
(w) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 9.90 W (max); 6.64 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
DII	NA	0.4	2.4	-6.03	-2.85	0.22	100
AD ID B09	-1.5	0.4	2.4	-0.36	0.02	0.1	75
AD ID B08	-1.5	0.4	2.4	-0.36	0.02	0.1	75
AD ID B07	-1.5	0.4	2.4	-0.36	0.02	0.1	75
AD ID B06	-1.5	0.4	2.4	-0.36	0.02	0.1	75
AD ID B05	-1.5	0.4	2.4	-0.36	0.02	0.1	75
AD ID B04	-1.5	0.4	2.4	-0.36	0.02	0.1	75
AD ID B03	-1.5	0.4	2.4	-0.36	0.02	0.1	75
AD ID B02	-1.5	0.4	2.4	-0.36	0.02	0.1	75
MMIO IRQ IN	NA	0.4	2.4	-6.63	-2.83	0.32	100
RD DAT 15	NA	0.4	2.4	-0.25	0.07	NA	NA
RD DAT 14	NA	0.4	2.4	-0.25	0.07	NA	NA
RD DAT 13	NA	0.4	2.4	-0.25	0.07	NA	NA

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
RD DAT 12	NA	0.4	2.4	-0.61	0.09	NA	NA
RD DAT 11	NA	0.4	2.4	-0.61	0.09	NA	NA
RD DAT 10	NA	0.4	2.4	-0.61	0.09	NA	NA
RD DAT 9	NA	0.4	2.4	-0.61	0.09	NA	NA
RD DAT 8	NA	0.4	2.4	-0.61	0.09	NA	NA
AD ID B12	-1.5	0.4	2.4	-0.36	0.02	0.1	75
AD ID B11	-1.5	0.4	2.4	-0.36	0.02	0.1	50
AD ID B10	-1.5	0.4	2.4	-0.36	0.02	0.1	75
WR LOWER	-1.2	0.4	2.4	-1.8	0.06	0.3	100
WR UPPER	-1.2	0.4	2.4	-1.8	0.06	0.3	100
LOAD EN	NA	0.4	2.4	-5.63	-2.85	0.22	75
RD DAT 7	NA	0.4	2.4	-0.61	0.09	NA	NA
RD DAT 6	NA	0.4	2.4	-0.61	0.09	NA	NA

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
RD DAT 5	NA	0.4	2.4	-0.61	0.09	NA	NA
RD DAT 4	NA	0.4	2.4	-0.61	0.09	NA	NA
RD DAT 3	NA	0.4	2.4	-0.61	0.09	NA	NA
RD DAT 2	NA	0.4	2.4	-0.61	0.09	NA	NA
RD DAT 1	NA	0.4	2.4	-0.85	0.09	NA	NA
RD DAT 0	NA	0.4	2.4	-0.85	0.09	NA	NA
EN INT DAT	-1.5	0.4	2.4	-0.4	0.04	0.2	125
EN CMP	-1.2	0.4	2.4	-1.8	0.06	0.3	100
MMIO B/IO SE	-1.2	0.4	2.4	-0.6	0.02	0.1	75
MMIO CMP SEL	-1.2	0.4	2.4	-0.6	0.02	0.1	75
MMIO MACLR	-1.5	0.4	2.4	-0.2	0.02	0.1	100
MMIO CLK	-1.2	0.4	2.4	-4.0	0.1	2.0	75
EXT ID B8	-1.5	0.4	2.4	-0.2	0.02	0.1	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
EXT ID B9	-1.5	0.4	2.4	-0.2	0.02	0.1	75
EXT ID B10	-1.5	0.4	2.4	-0.2	0.02	0.1	75
EXT ID B11	-1.5	0.4	2.4	-0.2	0.02	0.1	75
EXT ID B12	-1.5	0.4	2.4	-0.2	0.02	0.1	75
EXT ID B13	-1.5	0.4	2.4	-0.2	0.02	0.1	75
EXT ID B14	-1.5	0.4	2.4	-0.2	0.02	0.1	75
EXT ID B15	-1.5	0.4	2.4	-0.2	0.02	0.1	75
PAR/SER	NA	0.4	2.4	-6.63	-2.75	0.72	200
EXT IN LD1	-1.2	0.4	2.4	-0.6	0.02	0.1	75
EXT IN LD0	-1.2	0.4	2.4	-0.6	0.02	0.1	75
EXT OUT SEL1	-1.2	0.4	2.4	-0.6	0.02	0.1	75
EXT OUT SEL0	-1.2	0.4	2.4	-0.6	0.02	0.1	75
EXT ID B0	-1.5	0.4	2.4	-0.6	0.04	0.2	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
EXT ID B1	-1.5	0.4	2.4	-1.0	0.06	0.3	75
EXT ID B2	-1.5	0.4	2.4	-0.2	0.02	0.1	75
EXT ID B3	-1.5	0.4	2.4	-0.2	0.02	0.1	50
EXT ID B4	-1.5	0.4	2.4	-0.2	0.02	0.1	75
EXT ID B5	-1.5	0.4	2.4	-0.2	0.02	0.1	75
EXT ID B6	-1.5	0.4	2.4	-0.2	0.02	0.1	75
EXT ID B7	-1.5	0.4	2.4	-0.2	0.02	0.1	75
DEV ID 5	-1.5	0.4	2.4	-0.2	0.02	0.1	75
DEV ID 4	-1.5	0.4	2.4	-0.2	0.02	0.1	75
DEV ID 3	-1.5	0.4	2.4	-0.2	0.02	0.1	75
DEV ID 2	-1.5	0.4	2.4	-0.2	0.02	0.1	75
DEV ID 1	-1.5	0.4	2.4	-0.2	0.02	0.1	75
DEV ID 0	-1.5	0.4	2.4	-0.2	0.02	0.1	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
IIU	-1.5	0.4	2.4	-0.2	0.02	0.1	100
EXT OUT EN	NA	0.4	2.4	-6.03	-2.81	0.42	175
EXT OD B0	NA	NA	NA	-1.05	0.11	NA	NA

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
C/S RG B7	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
C/S RG B6	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
C/S RG B5	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
C/S RG B4	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
MMIO IRQ OUT	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
RD DAT 15	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
RD DAT 14	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
RD DAT 13	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
RD DAT 12	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
RD DAT 11	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
RD DAT 10	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
RD DAT 9	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
RD DAT 8	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
RD DAT 7	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
RD DAT 6	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
RD DAT 5	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
RD DAT 4	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
RD DAT 3	2.4	0.5	-0.4	6.0	NA	NA	-60.0	125
RD DAT 2	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
RD DAT 1	2.4	0.5	-0.4	6.0	NA	NA	-60.0	125
RD DAT 0	2.4	0.5	-0.4	6.0	NA	NA	-60.0	125
MMIO REQ	NA	0.5	NA	6.0	0.25	NA	NA	NA
MMIO PRES	NA	0.5	NA	6.0	0.25	NA	NA	NA
EXT OD B15	2.4	0.5	-0.4	6.0	0.05	0.05	-60.0	NA
EXT OD B14	2.4	0.5	-0.4	6.0	0.05	0.05	-60.0	NA
EXT OF B13	2.4	0.5	-0.4	6.0	0.05	0.05	-60.0	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
EXT OD B12	2.4	0.5	-0.4	6.0	0.05	0.05	-60.0	NA
EXT OD B11	2.4	0.5	-0.4	6.0	0.05	0.05	-60.0	NA
EXT OD B10	2.4	0.5	-0.4	6.0	0.05	0.05	-60.0	NA
EXT OD B9	2.4	0.5	-0.4	6.0	0.05	0.05	-60.0	NA
EXT OD B8	2.4	0.5	-0.4	6.0	0.05	0.05	-60.0	NA
EXT OD B7	2.4	0.5	-0.4	6.0	0.05	0.05	-60.0	NA
EXT OD B6	2.4	0.5	-0.4	6.0	0.05	0.05	-60.0	NA
EXT OD B5	2.4	0.5	-0.4	6.0	0.05	0.05	-60.0	NA
EXT OD B4	2.4	0.5	-0.4	6.0	0.05	0.05	-60.0	NA
EXT OD B3	2.4	0.5	-0.4	6.0	0.05	0.05	-60.0	NA
EXT OD B2	2.4	0.5	-0.4	6.0	0.05	0.05	-60.0	NA
EXT OD B1	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
EXT OD B0	2.4	0.5	-0.4	6.0	0.1	0.1	-60.0 -100.0	NA

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6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	MMIO CLK to MMIO REQ	-	-	-	37	37	37	nS	11
2	tpd	MMIO CLK to MMIO REQ	-	-	-	39	39	39	nS	11
3	tpd	MMIO CLK to C/S RG B5	-	-	-	66	66	66	nS	11
4	tpd	EXT OUT EN to C/S RG B5	-	-	-	55	55	55	nS	11
5	tpd	MMIO CLK to EXT OD B0	-	-	-	73	73	73	nS	11
5	tpd	MMIO CLK to EXT OD B1	-	-	-	73	73	73	nS	11
5	tpd	MMIO CLK to EXT OD B2	-	-	-	73	73	73	nS	11
5	tpd	MMIO CLK to EXT OD B3	-	-	-	73	73	73	nS	11
5	tpd	MMIO CLK to EXT OD B4	-	-	-	73	73	73	nS	11
5	tpd	MMIO CLK to EXT OD B5	-	-	-	73	73	73	nS	11
5	tpd	MMIO CLK to EXT OD B6	-	-	-	73	73	73	nS	11

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
5	tpd	MMIO CLK to EXT OD B7	-	-	-	73	73	73	nS	11
5	tpd	MMIO CLK to EXT OD B8	-	-	-	73	73	73	nS	11
5	tpd	MMIO CLK to EXT OD B9	-	-	-	73	73	73	nS	11
5	tpd	MMIO CLK to EXT OD B10	-	-	-	73	73	73	nS	11
5	tpd	MMIO CLK to EXT OD B11	-	-	-	73	73	73	nS	11
5	tpd	MMIO CLK to EXT OD B12	-	-	-	73	73	73	nS	11
5	tpd	MMIO CLK to EXT OD B13	-	-	-	73	73	73	nS	11
5	tpd	MMIO CLK to EXT OD B14	-	-	-	73	73	73	nS	11
5	tpd	MMIO CLK to EXT OD B15	-	-	-	73	73	73	nS	11
6	tpd	EXT OUT EN to EXT OD B0	-	-	-	30	30	30	nS	11
6	tpd	EXT OUT EN to EXT OD B1	-	-	-	30	30	30	nS	11
6	tpd	EXT OUT EN to EXT OD B2	-	-	-	30	30	30	nS	11

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
6	tpd	EXT OUT EN to EXT OD B3	-	-	-	30	30	30	nS	11
6	tpd	EXT OUT EN to EXT OD B4	-	-	-	30	30	30	nS	11
6	tpd	EXT OUT EN to EXT OD B5	-	-	-	30	30	30	nS	11
6	tpd	EXT OUT EN to EXT OD B6	-	-	-	30	30	30	nS	11
6	tpd	EXT OUT EN to EXT OD B7	-	-	-	30	30	30	nS	11
6	tpd	EXT OUT EN to EXT OD B8	-	-	-	30	30	30	nS	11
6	tpd	EXT OUT EN to EXT OD B9	-	-	-	30	30	30	nS	11
6	tpd	EXT OUT EN to EXT OD B10	-	-	-	30	30	30	nS	11
6	tpd	EXT OUT EN to EXT OD B11	-	-	-	30	30	30	nS	11
6	tpd	EXT OUT EN to EXT OD B12	-	-	-	30	30	30	nS	11
6	tpd	EXT OUT EN to EXT OD B13	-	-	-	30	30	30	nS	11
6	tpd	EXT OUT EN to EXT OD B14	-	-	-	30	30	30	nS	11

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
6	tpd	EXT OUT EN to EXT OD B15	-	-	-	30	30	30	nS	11
7 1/	tpd	MMIO CLK to EXT OD B8	-	-	-	73	73	73	nS	11
7	tpd	MMIO CLK to EXT OD B9	-	-	-	73	73	73	nS	11
7	tpd	MMIO CLK to EXT OD B10	-	-	-	73	73	73	nS	11
7	tpd	MMIO CLK to EXT OD B11	-	-	-	73	73	73	nS	11
7	tpd	MMIO CLK to EXT OD B12	-	-	-	73	73	73	nS	11
7	tpd	MMIO CLK to EXT OD B13	-	-	-	73	73	73	nS	11
7	tpd	MMIO CLK to EXT OD B14	-	-	-	73	73	73	nS	11
7	tpd	MMIO CLK to EXT OD B15	-	-	-	73	73	73	nS	11
8 1/	tpd	EXT OUT EN to EXT OD B8	-	-	-	30	30	30	nS	11
8	tpd	EXT OUT EN to EXT OD B9	-	-	-	30	30	30	nS	11
8	tpd	EXT OUT EN to EXT OD B10	-	-	-	30	30	30	nS	11

1/ For these tests pins 41, 42, 72, and 73 shall be set to 1.

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
8 1/	tpd	EXT OUT EN to EXT OD B11	-	-	-	30	30	30	nS	11
8	tpd	EXT OUT EN to EXT OD B12	-	-	-	30	30	30	nS	11
8	tpd	EXT OUT EN to EXT OD B13	-	-	-	30	30	30	nS	11
8	tpd	EXT OUT EN to EXT OD B14	-	-	-	30	30	30	nS	11
8	tpd	EXT OUT EN to EXT OD B15	-	-	-	30	30	30	nS	11
9	tpd	EXT OD B1 to EXT OD B0	-	-	-	53	53	53	nS	11
10	tpd	LOAD EN to C/S RG B7	-	-	-	33	33	33	nS	11
11	tpd	MMIO CLK to C/S RG B7	-	-	-	75	75	75	nS	11
12	tpd	LOAD EN to C/S RG B6	-	-	-	33	33	33	nS	11
13	tpd	MMIO CLK to C/S RG B6	-	-	-	75	75	75	nS	11
14	tpd	MMIO CLK to RD DAT 0	-	-	-	73	73	73	nS	11

1/ For these tests pins 41, 42, 72, and 73 shall be set to 1.

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
14	tpd	MMIO CLK to RD DAT 1	-	-	-	73	73	73	nS	11
14	tpd	MMIO CLK to RD DAT 2	-	-	-	73	73	73	nS	11
14	tpd	MMIO CLK to RD DAT 3	-	-	-	73	73	73	nS	11
14	tpd	MMIO CLK to RD DAT 4	-	-	-	73	73	73	nS	11
14	tpd	MMIO CLK to RD DAT 5	-	-	-	73	73	73	nS	11
14	tpd	MMIO CLK to RD DAT 6	-	-	-	73	73	73	nS	11
14	tpd	MMIO CLK to RD DAT 7	-	-	-	73	73	73	nS	11
14	tpd	MMIO CLK to RD DAT 8	-	-	-	73	73	73	nS	11
14	tpd	MMIO CLK to RD DAT 9	-	-	-	73	73	73	nS	11
14	tpd	MMIO CLK to RD DAT 10	-	-	-	73	73	73	nS	11
14	tpd	MMIO CLK to RD DAT 11	-	-	-	73	73	73	nS	11
14	tpd	MMIO CLK to RD DAT 12	-	-	-	73	73	73	nS	11

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
14	tpd	MMIO CLK to RD DAT 13	-	-	-	73	73	73	nS	11
14	tpd	MMIO CLK to RD DAT 14	-	-	-	73	73	73	nS	11
14	tpd	MMIO CLK to RD DAT 15	-	-	-	73	73	73	nS	11
15	tpd	MMIO CLK to RD DAT 0	-	-	-	73	73	73	nS	11
15	tpd	MMIO CLK to RD DAT 1	-	-	-	73	73	73	nS	11
15	tpd	MMIO CLK to RD DAT 2	-	-	-	73	73	73	nS	11
15	tpd	MMIO CLK to RD DAT 3	-	-	-	73	73	73	nS	11

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6.4 Module function. Memory Mapped Input/Output (MMIO) channels provide data transfers between the MRP and the external equipment (without using Input/Output Channel Adapters (IOCAs)), allowing the Central Processor (CP) to use the same instruction for Input/Output (I/O) transfers as it does for memory transfers. (The I/O device appears as a memory location to the CP or Input/Output Controller (IOC)). Up to 64 MMIO channels may be implemented in the MRP. Each MMIO channel operates independently and is contained on one module. An MMIO bus interface module (CWA) provides the interface between 1 to 16 MMIO channels and the MRP common bus. An MMIO channel contains four registers: output data, input data, external interrupt data, and control/status data. Data and status information are exchanged between the external equipment and the MRP (via a Transistor Transistor Logic (TTL) compatible interface) under control of the external equipment (at the rate and in the mode defined by the external equipment; either 16-bit parallel or serial). Each MMIO register has an assigned address within the first 8K of main memory (0000-1FFF) which the external equipment is required to access. Main memory addresses within this range are not used and will be inhibited when MMIO is referenced. Based upon the condition of the control/status register, the MMIO channel generates an interrupt to indicate the completion of a data transfer or the existence of an external equipment discrete interrupt condition. The interrupt code and the MMIO channel identification are presented to the CP over the common bus.

6.4.1 Memory mapped input/output adapter. The MMIO adapter module, hereafter referred to as MMIO, contains a TTL-compatible interface to an external device. The module contains input and output shift registers, an external interrupt register, an input data register, output data register, Control and Status (C/S) register, interface control logic, MMIO interrupt control logic, register selector, and address comparator.

6.4.1.1 Input and output shift registers. The input and output shift registers accomplish the parallel-to-serial conversion for an output transfer and serial-to-parallel conversion for an input transfer (serial channel transfers). The shift clock is output data bit 1 and input data bit 1 for output and input, respectively. Parallel output data will be sent directly to/from the external device via the external interface lines, bypassing the input and output shift registers. Selection of parallel or serial modes of operation is controlled by the interface control logic.

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6.4.1.2 Control and status register. The control and status register is a 16-bit register containing interrupt control bits and condition flags. Bits 4 through 7 and 12 through 15 are used for discrete functions which control the communication between an external device and the CP or IOC. Bits 4 and 8 through 15 are directly modifiable by a direct load from the CP or IOC using the assigned address. The function of these bits is as follows:

(a) Bits 7 and 15 - If bit 15 is set, it indicates to the external device that it may send an external interrupt with an associated External Interrupt (EI) word, in other words, the CP is prepared to receive a class III interrupt and an EI word and process it. The interrupt word is 16 bits in length. Bit 7 of the C/S register indicates that the interrupt word has been transferred to the MMIO channel. The CP will receive a class III interrupt, and upon reading the EI word, bit 7 will be cleared by the hardware.

(b) Bits 6 and 14 - These two bits work together similar to the EI C/S bits 7 and 15. If bit 14 is set, the CP or IOC indicate they are prepared to receive an input word. If bit 6 is set, the input word has been transferred to the input data register on the MMIO module by the external device and a class III interrupt will be requested. Bit 6 will be cleared upon a read of the input word.

(c) Bits 5 and 13 - If bit 13 is set, the CP or IOC indicate they are prepared to transfer data to the external device. Bit 5 is set when the CP or IOC loads the output data register. Bit 5 will be cleared as the external device reads the output register, and normally, a class III output data interrupt request will be initiated for subsequent output data.

6.4.1.3 Output data register. The output data register is a 16-bit data register with dual outputs. Data outputs are selected by the output decoder logic as a function of the external device supplied output code bits. Information outputs are enabled by the register decoder and are supplied as read data to the information bus.

6.4.1.4 Input data register. The input data register is a 16-bit data register with dual outputs. Register loading from the external interface lines is controlled by the load control bit decoder as a function of the external device supplied load code bits. Data will be enabled onto the information bus when addressed by the register decoder.

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6.4.1.5 External interrupt register. The external interrupt register is a 16-bit data register with dual outputs. Interrupt data out to the information bus is enabled by the register selector as a function of address/write data bits 1-0. Register loading is accomplished by an enable derived from the external device supplied load code bits.

6.4.1.6 Register decoder. The register decoder provides enables for the EI register, input and output data registers, and the C/S register. Outputs from these registers are placed onto the information bus. Address/write data bits 1-0 are decoded by the register selector as follows:

Address/write data Bit 1	Bit 0	Register selected
0	0	C/S register
0	1	Output data register
1	0	Input data register
1	1	EI data register

6.4.1.7 Address comparator. The address comparator logic is composed of two 8-bit comparators used to check the incoming address against the hard-wired device address assigned to the MMIO channel. Address compare is enabled by the Memory Mapped Bus Interface (MMBI) address detector logic and if a match occurs, the MMIO request signal is generated and sent to the memory interface module to ensure that the memory cycle is aborted.

6.4.1.8 MMIO interrupt control. Within a group of MMIO channels, behind an MMBI module, an interrupt line is daisy-chained in and out of consecutive MMIO modules and ends up driving an interrupt line to the MMBI module. The priority is designed to have the MMIO module on the end of the daisy-chain to have priority in the event of a simultaneous request. Bits 15-12 of the C/S register control the generation of interrupts on the MMIO module. Interrupts will be latched and sent to the priority encoder. Priority encoder outputs are interpreted as follows:

Interrupt code bits Bit 2	Bit 1	Function
0	0	External interrupt
0	1	Input data interrupt
1	0	Output data interrupt
1	1	Discrete interrupt indicator

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The external interrupt and input data interrupt imply the respective type word is available to be read from an MMIO channel. In general, the output data interrupt means the external device has read the output data register and prepared to receive the next output word. No data is associated with the Discrete Interrupt Indicator (DII). Priority encoder bits are sent to the MMIO interrupt register along with the hard-wired device Input Data (ID) code. The priority lock-out logic allows the higher priority (numbered) MMIO to lock out the lower numbered MMIO. The MMIO Intr Req Out signal will be passed onto the next MMIO or the controlling MMBI in the daisy-chain.

6.4.1.9 Interface control logic. The interface control logic consists of parallel and serial control logic and load and output control decoders. Load and output control decoders receive external device supplied load code 1-0 and out code 1-0 and supply load enables and output enables to EI, input data, output data, and control status registers. Parallel and serial control logic receives the parallel, serial and the load enable signals from the external device and generates the Serial Mode signal (control for serial input shift register), the Load Register signal (clock for the C/S, ID, and EI registers), the Out Serial Data signal (serial data output, Output Data (OD) bit 0), the Enable Parallel Out signal (output enable for output parallel data bus latches), and the MMIO clock.

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6.4.1.10 Description of I/O pins.

Signal name	Pin number	I/O	Description
DII	2	I	DII (DISCRETE INTRPT IND)
C/S RG B7	3	O	C/S REG BIT 7
C/S RG B6	4	O	C/S REG BIT 6
C/S RG B5	5	O	C/S REG BIT 5
C/S RG B4	6	O	C/S REG BIT 4
AD ID B09	7	I	ADRS ID BIT 09
AD ID B08	8	I	ADRS ID BIT 08
AD ID B07	9	I	ADRS ID BIT 07
AD ID B06	11	I	ADRS ID BIT 06
AD ID B05	12	I	ADRS ID BIT 05
AD ID B04	13	I	ADRS ID BIT 04
AD ID B03	14	I	ADRS ID BIT 03
AD ID B02	15	I	ADRS ID BIT 02
MMIO IRQ IN	16	I	MMIO INTRPT REQ IN
MMIO IRQ OUT	17	O	MMIO INTRPT REQ OUT
RD DAT 15	18	I/O	MMIO ADRS/DATA 15
RD DAT 14	19	I/O	MMIO ADRS/DATA 14
RD DAT 13	20	I/O	MMIO ADRS/DATA 13
RD DAT 12	21	I/O	MMIO ADRS/DATA 12
RD DAT 11	22	I/O	MMIO ADRS/DATA 11

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6.4.1.10 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
RD DAT 10	23	I/O	MMIO ADRS/DATA 10
RD DAT 9	24	I/O	MMIO ADRS/DATA 09
RD DAT 8	26	I/O	MMIO ADRS/DATA 08
AD ID B12	27	I	ADRS ID BIT 12
AD ID B11	28	I	ADRS ID BIT 11
AD ID B10	29	I	ADRS ID BIT 10
WR LOWER	30	I	WR LOWER
WR UPPER	32	I	WR UPPER
LOAD EN	33	I	LOAD EN
RD DAT 7	34	I/O	MMIO ADRS/DATA 07
RD DAT 6	35	I/O	MMIO ADRS/DATA 06
RD DAT 5	36	I/O	MMIO ADRS/DATA 05
RD DAT 4	37	I/O	MMIO ADRS/DATA 04
RD DAT 3	38	I/O	MMIO ADRS/DATA 03
RD DAT 2	39	I/O	MMIO ADRS/DATA 02
RD DAT 1	41	I/O	MMIO ADRS/DATA 01
RD DAT 0	42	I/O	MMIO ADRS/DATA 00
EN INT DAT	43	I	ENBL INTRPT DATA
MMIO REQ	44	O	MMIO REQ
EN CMP	45	I	ENABLE COMPARE

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6.4.1.10 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
MMIO PRES	46	O	MMIO PRESENT
MMIO B/ID SE	47	I	MMIO BITE/ID SEL
MMIO CMP SEL	48	I	MMIO COMPARE SEL
MMIO MACLR	49	I	MMIO MACLR
MMIO CLK	50	I	MMIO CLK (20 MHz)
EXT ID B8	52	I	EXT ID BIT 08
EXT ID B9	53	I	EXT ID BIT 09
EXT ID B10	54	I	EXT ID BIT 10
EXT ID B11	55	I	EXT ID BIT 11
EXT ID B12	56	I	EXT ID BIT 12
EXT ID B13	57	I	EXT ID BIT 13
EXT ID B14	58	I	EXT ID BIT 14
EXT ID B15	59	I	EXT ID BIT 15
EXT OD B15	61	O	EXT OD BIT 15
EXT OD B14	62	O	EXT OD BIT 14
EXT OD B13	63	O	EXT OD BIT 13
EXT OD B12	64	O	EXT OD BIT 12
EXT OD B11	65	O	EXT OD BIT 11
EXT OD B10	66	O	EXT OD BIT 10
EXT OD B9	67	O	EXT OD BIT 09

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6.4.1.10 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
EXT OD B8	68	O	EXT OD BIT 08
PAR/SER	69	I	PARALLEL/SERIAL
EXT IN LD1	70	I	EXT INPUT LD CODE 1
EXT IN LD0	71	I	EXT INPUT LD CODE 0
EXT OUT SEL1	72	I	EXT OUTPUT SEL CODE 1
EXT OUT SEL0	73	I	EXT OUTPUT SEL CODE 0
EXT ID B0	74	I	EXT ID BIT 00
EXT ID B1	76	I	EXT ID BIT 01 + SERIAL CLK
EXT ID B2	77	I	EXT ID BIT 02
EXT ID B3	78	I	EXT ID BIT 03
EXT ID B4	79	I	EXT ID BIT 04
EXT ID B5	80	I	EXT ID BIT 05
EXT ID B6	82	I	EXT ID BIT 06
EXT ID B7	83	I	EXT ID BIT 07
DEV ID 5	84	I	DEVICE ID 5
DEV ID 4	85	I	DEVICE ID 4
DEV ID 3	86	I	DEVICE ID 3
DEV ID 2	87	I	DEVICE ID 2
DEV ID 1	88	I	DEVICE ID 1
DEV ID 0	89	I	DEVICE ID 0

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6.4.1.10 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
IIU	91	I	INHIBIT INTERRUPT UPDATE
EXT OD B7	92	O	EXT OD BIT 07
EXT OD B6	93	O	EXT OD BIT 06
EXT OD B5	94	O	EXT OD BIT 04
EXT OD B4	95	O	EXT OD BIT 04
EXT OD B3	96	O	EXT OD BIT 03
EXT OD B2	97	O	EXT OD BIT 02
EXT OD B1	98	I/O	EXT OD BIT 01 + SERIAL CLOCK
EXT OD B0	99	O	EXT OD BIT 00
EXT OUT EN	100	I	EXT OUTPUT ENBL

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6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT, AV
Navy - AS, MC
Air Force - 13, 17, 19
DLA - ES

Preparing activity:
Navy - SH

(Project 5963-0044)

Agent:
NWSC Crane

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TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A	Isolation, frame	C	Frame	10	INF	MOhm	1-8

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A
Isolation measurement	NA	± 500 KOhm	A

2. The ohmmeter used shall be item R06.

3. In the set-up conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. Caution: Set megohm bridge to limit current to 10 uA before applying voltage to the module.

7. The test circuit shall be as indicated on figure 6.

8. Abbreviations used are defined in 6.1.

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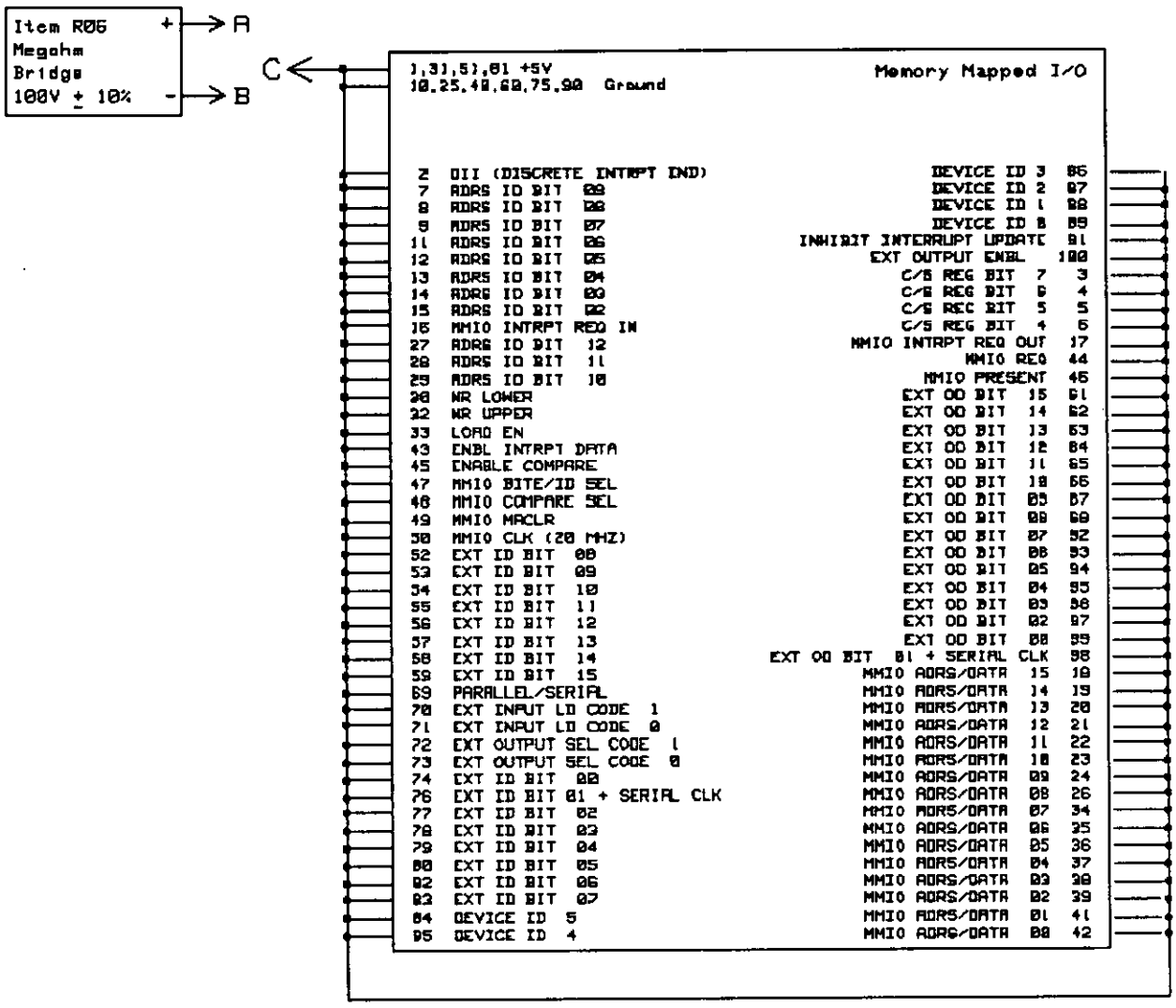


FIGURE 6. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, GND	10	25	0	1	Ohm	1-5
B2	Continuity, GND	10	40				
B3	Continuity, GND	60	75				
B4	Continuity, GND	60	90				
B5	Continuity, Vcc	1	31				
B6	Continuity, Vcc	51	81	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B6

2. The ohmmeter used shall be item R07.

3. In the set-up conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

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1,21,51,81 +5V			Memory Mapped I/O
10,25,40,60,75,90 Ground			
2	DII (DISCRETE INTRPT IND)	DEVICE ID 3	86
7	ADRS IO BIT 09	DEVICE ID 2	87
8	ADRS IO BIT 08	DEVICE ID 1	88
9	ADRS IO BIT 07	DEVICE ID 0	89
11	ADRS IO BIT 06	INHIBIT INTERRUPT UPDATE	91
12	ADRS IO BIT 05	EXT OUTPUT ENBL	100
13	ADRS IO BIT 04	C/S REG BIT 7	3
14	ADRS IO BIT 03	C/S REG BIT 6	4
15	ADRS IO BIT 02	C/S REG BIT 5	5
16	MMIO INTRPT REQ IN	C/S REG BIT 4	6
27	ADRS IO BIT 12	MMIO INTRPT REQ OUT	17
28	ADRS IO BIT 11	MMIO REQ	44
29	ADRS IO BIT 10	MMIO PRESENT	46
30	WR LOWER	EXT OD BIT 15	61
32	WR UPPER	EXT OD BIT 14	62
33	LOAD EN	EXT OD BIT 13	63
43	ENBL INTRPT DATA	EXT OD BIT 12	64
45	ENABLE COMPARE	EXT OD BIT 11	65
47	MMIO BITE/ID SEL	EXT OD BIT 10	66
48	MMIO COMPARE SEL	EXT OD BIT 09	67
49	MMIO MRCLR	EXT OD BIT 08	68
50	MMIO CLK (20 MHZ)	EXT OD BIT 07	92
52	EXT ID BIT 08	EXT OD BIT 06	93
53	EXT ID BIT 09	EXT OD BIT 05	94
54	EXT ID BIT 10	EXT OD BIT 04	95
55	EXT ID BIT 11	EXT OD BIT 03	96
56	EXT ID BIT 12	EXT OD BIT 02	97
57	EXT ID BIT 13	EXT OD BIT 00	99
58	EXT ID BIT 14	EXT OD BIT 01 + SERIAL CLK	98
59	EXT ID BIT 15	MMIO ADRS/DATA 15	18
69	PARALLEL/SERIAL	MMIO ADRS/DATA 14	19
70	EXT INPUT LD CODE 1	MMIO ADRS/DATA 13	20
71	EXT INPUT LD CODE 0	MMIO ADRS/DATA 12	21
72	EXT OUTPUT SEL CODE 1	MMIO ADRS/DATA 11	22
73	EXT OUTPUT SEL CODE 0	MMIO ADRS/DATA 10	23
74	EXT ID BIT 00	MMIO ADRS/DATA 09	24
76	EXT ID BIT 01 + SERIAL CLK	MMIO ADRS/DATA 08	26
77	EXT ID BIT 02	MMIO ADRS/DATA 07	34
78	EXT ID BIT 03	MMIO ADRS/DATA 06	35
79	EXT ID BIT 04	MMIO ADRS/DATA 05	36
80	EXT ID BIT 05	MMIO ADRS/DATA 04	37
82	EXT ID BIT 06	MMIO ADRS/DATA 03	38
83	EXT ID BIT 07	MMIO ADRS/DATA 02	39
84	DEVICE ID 5	MMIO ADRS/DATA 01	41
85	DEVICE ID 4	MMIO ADRS/DATA 00	42

FIGURE 7. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1,31,51, 81	10,25, 40,60, 75,90	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	3.24 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

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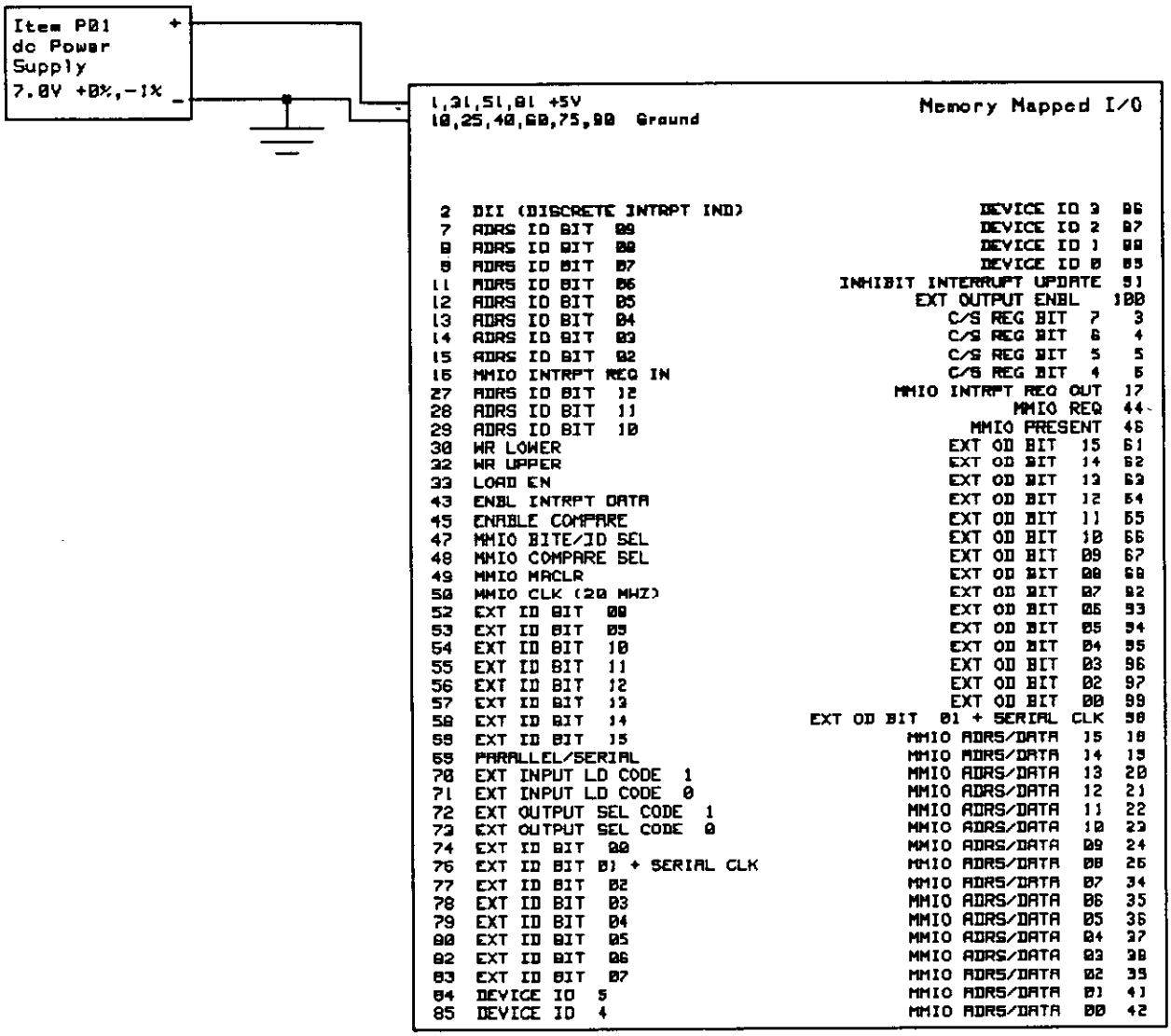


FIGURE 8. Vcc stress test circuit for table IV.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	DII	2	5.5	-0.12	0.22	mA	1-3
D2	AD ID B09	7	5.5	0.0	0.1	mA	1-3
D3	AD ID B08	8					
D4	AD ID B07	9					
D5	AD ID B06	11					
D6	AD ID B05	12					
D7	AD ID B04	13					
D8	AD ID B03	14					
D9	AD ID B02	15	5.5	0.0	0.1	mA	1-3
D10	MMIO IRQ IN	16	5.5	-0.12	0.32	mA	1-3
D11	AD ID B12	27	5.5	0.0	0.1	mA	1-3
D12	AD ID B11	28					
D13	AD ID B10	29	5.5	0.0	0.1	mA	1-3
D14	WR LOWER	30	5.5	0.0	0.3	mA	1-3
D15	WR UPPER	32	5.5	0.0	0.3	mA	1-3
D16	LOAD EN	33	5.5	-0.12	0.22	mA	1-3
D17	EN INT DAT	43	5.5	0.0	0.2	mA	1-3
D18	EN CMP	45	5.5	0.0	0.3	mA	1-3
D19	MMIO B/ID SE	47	5.5	0.0	0.1	mA	1-3
D20	MMIO CMP SEL	48	5.5	0.0	0.1	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
D21	MMIO MACLR	49	5.5	0.0	0.1	mA	1-3
D22	MMIO CLK	50	5.5	0.0	2.0	mA	1-3
D23	EXT ID B8	52	5.5	0.0	0.1	mA	1-3
D24	EXT ID B9	53					
D25	EXT ID B10	54					
D26	EXT ID B11	55					
D27	EXT ID B12	56					
D28	EXT ID B13	57					
D29	EXT ID B14	58					
D30	EXT ID B15	59	5.5	0.0	0.1	mA	1-3
D31	PAR/SER	69	5.5	-0.12	0.72	mA	1-3
D32	EXT IN LD1	70	5.5	0.0	0.1	mA	1-3
D33	EXT IN LD0	71					
D34	EXT OUT SEL1	72					
D35	EXT OUT SEL0	73	5.5	0.0	0.1	mA	1-3
D36	EXT ID B0	74	5.5	0.0	0.2	mA	1-3
D37	EXT ID B1	76	5.5	0.0	0.3	mA	1-3
D38	EXT ID B2	77	5.5	0.0	0.1	mA	1-3
D39	EXT ID B3	78					
D40	EXT ID B4	79	5.5	0.0	0.1	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D41	EXT ID B5	80	5.5	0.0	0.1	mA	1-3
D42	EXT ID B6	82					
D43	EXT ID B7	83					
D44	DEV ID 5	84					
D45	DEV ID 4	85					
D46	DEV ID 3	86					
D47	DEV ID 2	87					
D48	DEV ID 1	88					
D49	DEV ID 0	89					
D50	IIU	91	5.5	0.0	0.1	mA	1-3
D51	EXT OUT EN	100	5.5	-0.12	0.42	mA	1-3

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NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D51
Power supply current limit	2.7 A	NA	D1-D51
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D51
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 D1-D51 D1-D51
Input current measurement	NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA	± 17 uA ± 5 uA ± 22 uA ± 5 uA ± 15 uA ± 17 uA ± 10 uA ± 15 uA ± 5 uA ± 100 uA ± 5 uA ± 42 uA ± 5 uA ± 10 uA ± 15 uA ± 5 uA ± 27 uA	D1 D2-D9 D10 D11-D13 D14,D15 D16 D17 D18 D19-D21 D22 D23-D30 D31 D32-D35 D36 D37 D38-D50 D51

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	C/S RG B7	3	0.0	-150.0	-60.0	mA	1-4
E2	C/S RG B6	4					
E3	C/S RG B5	5					
E4	C/S RG B4	6					
E5	MMIO IRQ OUT	17					
E6	RD DAT 15	18					
E7	RD DAT 14	19					
E8	RD DAT 13	20					
E9	RD DAT 12	21					
E10	RD DAT 11	22					
E11	RD DAT 10	23					
E12	RD DAT 9	24					
E13	RD DAT 8	26					
E14	RD DAT 7	34					
E15	RD DAT 6	35					
E16	RD DAT 5	36					
E17	RD DAT 4	37					
E18	RD DAT 3	38					
E19	RD DAT 2	39					
E20	RD DAT 1	41	0.0	-150.0	-60.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Initial limits Max	Units	Notes
E21	RD DAT 0	42	0.0	-150.0	-60.0	mA	1-4
E22	EXT OD B15	61					
E23	EXT OD B14	62					
E24	EXT OD B13	63					
E25	EXT OD B12	64					
E26	EXT OD B11	65					
E27	EXT OD B10	66					
E28	EXT OD B9	67					
E29	EXT OD B8	68					
E30	EXT OD B7	92					
E31	EXT OD B6	93					
E32	EXT OD B5	94					
E33	EXT OD B4	95					
E34	EXT OD B3	96					
E35	EXT OD B2	97					
E36	EXT OD B1	98					
E37	EXT OD B0	99	0.0	-150.0	-60.0	mA	1-4
E38	EXT OD B0	99	0.0	-225.0	-100.0	mA	1-4

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NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E38
Power supply current limit	2.7 A	NA	E1-E38
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E38
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 E1-E38 E1-E38
Output current measurement	 NA NA	 ± 4.5 mA ± 6.25 mA	 E1-E37 E38

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	DII	2	2.4	-6.03	-2.85	mA	1-3
F2	AD ID B09	7	2.4	-0.36	0.02	mA	1-3
F3	AD ID B08	8					
F4	AD ID B07	9					
F5	AD ID B06	11					
F6	AD ID B05	12					
F7	AD ID B04	13					
F8	AD ID B03	14					
F9	AD ID B02	15	2.4	-0.36	0.02	mA	1-3
F10	MMIO IRQ IN	16	2.4	-6.63	-2.83	mA	1-3
F11	RD DAT 15	18	2.4	-0.25	0.07	mA	1-3
F12	RD DAT 14	19					
F13	RD DAT 13	20	2.4	-0.25	0.07	mA	1-3
F14	RD DAT 12	21	2.4	-0.61	0.09	mA	1-3
F15	RD DAT 11	22					
F16	RD DAT 10	23					
F17	RD DAT 9	24					
F18	RD DAT 8	26	2.4	-0.61	0.09	mA	1-3
F19	AD ID B12	27	2.4	-0.36	0.02	mA	1-3
F20	AD ID B11	28	2.4	-0.36	0.02	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F21	AD ID B10	29	2.4	-0.36	0.02	mA	1-3
F22	WR LOWER	30	2.4	-1.8	0.06	mA	1-3
F23	WR UPPER	32	2.4	-1.8	0.06	mA	1-3
F24	LOAD EN	33	2.4	-5.63	-2.85	mA	1-3
F25	RD DAT 7	34	2.4	-0.61	0.09	mA	1-3
F26	RD DAT 6	35					
F27	RD DAT 5	36					
F28	RD DAT 4	37					
F29	RD DAT 3	38					
F30	RD DAT 2	39	2.4	-0.61	0.09	mA	1-3
F31	RD DAT 1	41	2.4	-0.85	0.09	mA	1-3
F32	RD DAT 0	42	2.4	-0.85	0.09	mA	1-3
F33	EN INT DAT	43	2.4	-0.4	0.04	mA	1-3
F34	EN CMP	45	2.4	-1.8	0.06	mA	1-3
F35	MMIO B/ID SE	47	2.4	-0.6	0.02	mA	1-3
F36	MMIO CMP SEL	48	2.4	-0.6	0.02	mA	1-3
F37	MMIO MACLR	49	2.4	-0.2	0.02	mA	1-3
F38	MMIO CLK	50	2.4	-4.0	0.1	mA	1-3
F39	EXT ID B8	52	2.4	-0.2	0.02	mA	1-3
F40	EXT ID B9	53	2.4	-0.2	0.02	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F41	EXT ID B10	54	2.4	-0.2	0.02	mA	1-3
F42	EXT ID B11	55					
F43	EXT ID B12	56					
F44	EXT ID B13	57					
F45	EXT ID B14	58					
F46	EXT ID B15	59	2.4	-0.2	0.02	mA	1-3
F47	PAR/SER	69	2.4	-6.63	-2.75	mA	1-3
F48	EXT IN LD1	70	2.4	-0.6	0.02	mA	1-3
F49	EXT IN LD0	71					
F50	EXT OUT SEL1	72					
F51	EXT OUT SEL0	73	2.4	-0.6	0.02	mA	1-3
F52	EXT ID B0	74	2.4	-0.6	0.04	mA	1-3
F53	EXT ID B1	76	2.4	-1.0	0.06	mA	1-3
F54	EXT ID B2	77	2.4	-0.2	0.02	mA	1-3
F55	EXT ID B3	78					
F56	EXT ID B4	79					
F57	EXT ID B5	80					
F58	EXT ID B6	82					
F59	EXT ID B7	83					
F60	DEV ID 5	84	2.4	-0.2	0.02	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F61	DEV ID 4	85	2.4	-0.2	0.02	mA	1-3
F62	DEV ID 3	86					
F63	DEV ID 2	87					
F64	DEV ID 1	88					
F65	DEV ID 0	89					
F66	IIU	91	2.4	-0.2	0.02	mA	1-3
F67	EXT OD B1	98	2.4	-1.05	0.11	mA	1-3
F68	EXT OUT EN	100	2.4	-6.03	-2.81	mA	1-3

NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F68
Power supply current limit	2.7 A	NA	F1-F68
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F68
Pattern voltage high, (VH)	3.0 V	± 100 mV	F1-F68
low, (VL)	0.0 V	± 100 mV	F1-F68

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 159 uA	F1
	NA	± 19 uA	F2-F9
	NA	± 190 uA	F10
	NA	± 16 uA	F11-F13
	NA	± 35 uA	F14-F18
	NA	± 19 uA	F19-F21
	NA	± 93 uA	F22,F23
	NA	± 139 uA	F24
	NA	± 35 uA	F25-F30
	NA	± 47 uA	F31,F32
	NA	± 22 uA	F33
	NA	± 93 uA	F34
	NA	± 31 uA	F35,F36
	NA	± 11 uA	F37
	NA	± 205 uA	F38
	NA	± 11 uA	F39-F46
	NA	± 194 uA	F47
	NA	± 31 uA	F48-F51
	NA	± 32 uA	F52
	NA	± 53 uA	F53
	NA	± 11 uA	F54-F66
	NA	± 58 uA	F67
	NA	± 161 uA	F68

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	DII	2	0.4	-6.03	-2.85	mA	1-3
G2	AD ID B09	7	0.4	-0.36	0.02	mA	1-3
G3	AD ID B08	8					
G4	AD ID B07	9					
G5	AD ID B06	11					
G6	AD ID B05	12					
G7	AD ID B04	13					
G8	AD ID B03	14					
G9	AD ID B02	15	0.4	-0.36	0.02	mA	1-3
G10	MMIO IRQ IN	16	0.4	-6.63	-2.83	mA	1-3
G11	RD DAT 15	18	0.4	-0.25	0.07	mA	1-3
G12	RD DAT 14	19					
G13	RD DAT 13	20	0.4	-0.25	0.07	mA	1-3
G14	RD DAT 12	21	0.4	-0.61	0.09	mA	1-3
G15	RD DAT 11	22					
G16	RD DAT 10	23					
G17	RD DAT 9	24					
G18	RD DAT 8	26	0.4	-0.61	0.09	mA	1-3
G19	AD ID B12	27	0.4	-0.36	0.02	mA	1-3
G20	AD ID B11	28	0.4	-0.36	0.02	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G21	AD ID B10	29	0.4	-0.36	0.02	mA	1-3
G22	WR LOWER	30	0.4	-1.8	0.06	mA	1-3
G23	WR UPPER	32	0.4	-1.8	0.06	mA	1-3
G24	LOAD EN	33	0.4	-5.63	-2.85	mA	1-3
G25	RD DAT 7	34	0.4	-0.61	0.09	mA	1-3
G26	RD DAT 6	35					
G27	RD DAT 5	36					
G28	RD DAT 4	37					
G29	RD DAT 3	38					
G30	RD DAT 2	39	0.4	-0.61	0.09	mA	1-3
G31	RD DAT 1	41	0.4	-0.85	0.09	mA	1-3
G32	RD DAT 0	42	0.4	-0.85	0.09	mA	1-3
G33	EN INT DAT	43	0.4	-0.4	0.04	mA	1-3
G34	EN CMP	45	0.4	-1.8	0.06	mA	1-3
G35	MMIO B/ID SE	47	0.4	-0.6	0.02	mA	1-3
G36	MMIO CMP SEL	48	0.4	-0.6	0.02	mA	1-3
G37	MMIO MACLR	49	0.4	-0.2	0.02	mA	1-3
G38	MMIO CLK	50	0.4	-4.0	0.1	mA	1-3
G39	EXT ID B8	52	0.4	-0.2	0.02	mA	1-3
G40	EXT ID B9	53	0.4	-0.2	0.02	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G41	EXT ID B10	54	0.4	-0.2	0.02	mA	1-3
G42	EXT ID B11	55					
G43	EXT ID B12	56					
G44	EXT ID B13	57					
G45	EXT ID B14	58					
G46	EXT ID B15	59	0.4	-0.2	0.02	mA	1-3
G47	PAR/SER	69	0.4	-6.63	-2.75	mA	1-3
G48	EXT IN LD1	70	0.4	-0.6	0.02	mA	1-3
G49	EXT IN LD0	71					
G50	EXT OUT SEL1	72					
G51	EXT OUT SEL0	73	0.4	-0.6	0.02	mA	1-3
G52	EXT ID B0	74	0.4	-0.6	0.04	mA	1-3
G53	EXT ID B1	76	0.4	-1.0	0.06	mA	1-3
G54	EXT ID B2	77	0.4	-0.2	0.02	mA	1-3
G55	EXT ID B3	78					
G56	EXT ID B4	79					
G57	EXT ID B5	80					
G58	EXT ID B6	82					
G59	EXT ID B7	83					
G60	DEV ID 5	84	0.4	-0.2	0.02	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
G61	DEV ID 4	85	0.4	-0.2	0.02	mA	1-3
G62	DEV ID 3	86					
G63	DEV ID 2	87					
G64	DEV ID 1	88					
G65	DEV ID 0	89					
G66	IIU	91	0.4	-0.2	0.02	mA	1-3
G67	EXT OD B1	98	0.4	-1.05	0.11	mA	1-3
G68	EXT OUT EN	100	0.4	-6.03	-2.81	mA	1-3

NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G68
Power supply current limit	2.7 A	NA	G1-G68
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G68
Pattern voltage high, (VH)	3.0 V	± 100 mV	G1-G68
Pattern voltage low, (VL)	0.0 V	± 100 mV	G1-G68

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 159 uA	G1
	NA	± 19 uA	G2-G9
	NA	± 190 uA	G10
	NA	± 16 uA	G11-G13
	NA	± 35 uA	G14-G18
	NA	± 19 uA	G19-G21
	NA	± 93 uA	G22,G23
	NA	± 139 uA	G24
	NA	± 35 uA	G25-G30
	NA	± 47 uA	G31,G32
	NA	± 22 uA	G33
	NA	± 93 uA	G34
	NA	± 31 uA	G35,G36
	NA	± 11 uA	G37
	NA	± 205 uA	G38
	NA	± 11 uA	G39-G46
	NA	± 194 uA	G47
	NA	± 31 uA	G48-G51
	NA	± 32 uA	G52
	NA	± 53 uA	G53
	NA	± 11 uA	G54-G66
	NA	± 58 uA	G67
	NA	± 161 uA	G68

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H1	C/S RG B7	3	-0.4	2.4	4.5	V	1-4
H2	C/S RG B6	4					
H3	C/S RG B5	5					
H4	C/S RG B4	6					
H5	MMIO IRQ OUT	17					
H6	RD DAT 15	18					
H7	RD DAT 14	19					
H8	RD DAT 13	20					
H9	RD DAT 12	21					
H10	RD DAT 11	22					
H11	RD DAT 10	23					
H12	RD DAT 9	24					
H13	RD DAT 8	26					
H14	RD DAT 7	34					
H15	RD DAT 6	35					
H16	RD DAT 5	36					
H17	RD DAT 4	37					
H18	RD DAT 3	38					
H19	RD DAT 2	39					
H20	RD DAT 1	41	-0.4	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H21	RD DAT 0	42	-0.4	2.4	4.5	V	1-4
H22	EXT OD B15	61					
H23	EXT OD B14	62					
H24	EXT OD B13	63					
H25	EXT OD B12	64					
H26	EXT OD B11	65					
H27	EXT OD B10	66					
H28	EXT OD B9	67					
H29	EXT OD B8	68					
H30	EXT OD B7	92					
H31	EXT OD B6	93					
H32	EXT OD B5	94					
H33	EXT OD B4	95					
H34	EXT OD B3	96					
H35	EXT OD B2	97					
H36	EXT OD B1	98					
H37	EXT OD B0	99					
H38	EXT OD B0	99	-0.4	2.4	4.5	V	1-4

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NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	H1-H38
Power supply current limit	2.7 A	NA	H1-H38
Forcing current, (Ioh)	-0.4 mA	± 40 uA	H1-H38
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	H1-H38 H1-H38
Output voltage measurement	NA	± 105 mV	H1-H38

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

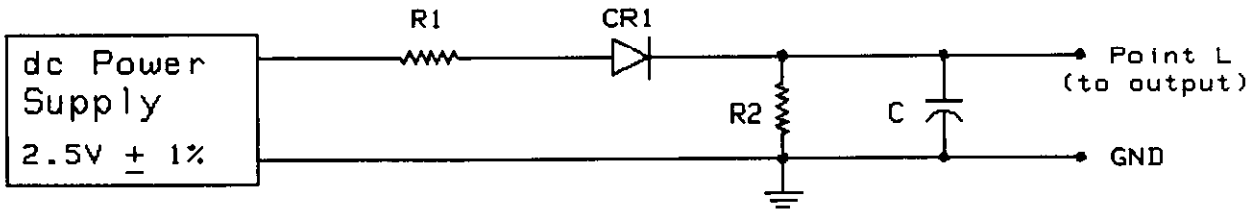
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

FIGURE 9. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
3,4,5,6,17,18,19, 20,21,22,23,24,26, 34,35,36,37,38,39, 41,42,61,62,63,64, 65,66,67,68,92,93, 94,95,96,97,98,99	6200	6.0	0.5
44,46	NA	6.0	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE X. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J1	C/S RG B7	3	6.0	0.0	0.5	V	1-4
J2	C/S RG B6	4					
J3	C/S RG B5	5					
J4	C/S RG B4	6					
J5	MMIO IRQ OUT	17					
J6	RD DAT 15	18					
J7	RD DAT 14	19					
J8	RD DAT 13	20					
J9	RD DAT 12	21					
J10	RD DAT 11	22					
J11	RD DAT 10	23					
J12	RD DAT 9	24					
J13	RD DAT 8	26					
J14	RD DAT 7	34					
J15	RD DAT 6	35					
J16	RD DAT 5	36					
J17	RD DAT 4	37					
J18	RD DAT 3	38					
J19	RD DAT 2	39					
J20	RD DAT 1	41	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J21	RD DAT 0	42	6.0	0.0	0.5	V	1-4
J22	MMIO REQ	44					
J23	MMIO PRES	46					
J24	MMIO PRES	46					
J25	MMIO PRES	46					
J26	EXT OD B15	61					
J27	EXT OD B14	62					
J28	EXT OD B13	63					
J29	EXT OD B12	64					
J30	EXT OD B11	65					
J31	EXT OD B10	66					
J32	EXT OD B9	67					
J33	EXT OD B8	68					
J34	EXT OD B7	92					
J35	EXT OD B6	93					
J36	EXT OD B5	94					
J37	EXT OD B4	95					
J38	EXT OD B3	96					
J39	EXT OD B2	97					
J40	EXT OD B1	98	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J41	EXT OD B0	99	6.0	0.0	0.5	V	1-4
J42	EXT OD B0	99	6.0	0.0	0.5	V	1-4

NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J42
Power supply current limit	2.7 A	NA	J1-J42
Forcing current, (Iol)	6.0 mA	± 600 uA	J1-J42
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	J1-J42 J1-J42
Output voltage measurement	NA	± 25 mV	J1-J42

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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TABLE XI. Output leakage current, high and low level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
High level							
K1	MMIO REQ	44	5.5	0.0	0.25	mA	1-3
K2	MMIO PRES	46	5.5	0.0	0.25	mA	1-3
K3	EXT OD B15	61	2.4	-0.05	0.05	mA	1-3
K4	EXT OD B14	62					
K5	EXT OD B13	63					
K6	EXT OD B12	64					
K7	EXT OD B11	65					
K8	EXT OD B10	66					
K9	EXT OD B9	67					
K10	EXT OD B8	68					
K11	EXT OD B7	92					
K12	EXT OD B6	93					
K13	EXT OD B5	94					
K14	EXT OD B4	95					
K15	EXT OD B3	96					
K16	EXT OD B2	97	2.4	-0.05	0.05	mA	1-3
K17	EXT OD B0	99	2.4	-0.1	0.1	mA	1-3

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TABLE XI. Output leakage current, high and low level, test - Continued.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
Low level							
K18	EXT OD B15	61	0.4	-0.05	0.05	mA	1-3
K19	EXT OD B14	62					
K20	EXT OD B13	63					
K21	EXT OD B12	64					
K22	EXT OD B11	65					
K23	EXT OD B10	66					
K24	EXT OD B9	67					
K25	EXT OD B8	68					
K26	EXT OD B7	92					
K27	EXT OD B6	93					
K28	EXT OD B5	94					
K29	EXT OD B4	95					
K30	EXT OD B3	96					
K31	EXT OD B2	97	0.4	-0.05	0.05	mA	1-3
K32	EXT OD B0	99	0.4	-0.1	0.1	mA	1-3

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NOTES:

1. Output leakage current, high and low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	K1-K32
Power supply current limit	2.7 A	NA	K1-K32
Forcing voltage,			
(Voh)	5.5 V	± 50 mV	K1,K2
(Voh)	2.4 V	± 50 mV	K3-K17
(Vol)	0.4 V	± 50 mV	K18-K32
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	K1-K32
low, (VL)	0.0 V	± 100 mV	K1-K32
Output current measurement	NA	± 12.5 uA	K1,K2
	NA	± 5.00 uA	K3-K16
	NA	± 10.0 uA	K17
	NA	± 5.00 uA	K18-K31
	NA	± 10.0 uA	K32

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XI and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Initial limits Max	Units	Notes
L1	AD ID B09	7	-18.0	-1.5	0.0	V	1-3
L2	AD ID B08	8					
L3	AD ID B07	9					
L4	AD ID B06	11					
L5	AD ID B05	12					
L6	AD ID B04	13					
L7	AD ID B03	14					
L8	AD ID B02	15					
L9	AD ID B12	27					
L10	AD ID B11	28					
L11	AD ID B10	29	-18.0	-1.5	0.0	V	1-3
L12	WR LOWER	30	-18.0	-1.2	0.0	V	1-3
L13	WR UPPER	32	-18.0	-1.2	0.0	V	1-3
L14	EN INT DAT	43	-18.0	-1.5	0.0	V	1-3
L15	EN CMP	45	-18.0	-1.2	0.0	V	1-3
L16	MMIO B/ID SE	47					
L17	MMIO CMP SEL	48	-18.0	-1.2	0.0	V	1-3
L18	MMIO MACLR	49	-18.0	-1.5	0.0	V	1-3
L19	MMIO CLK	50	-18.0	-1.2	0.0	V	1-3
L20	EXT ID B8	52	-18.0	-1.5	0.0	V	1-3

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TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L21	EXT ID B9	53	-18.0	-1.5	0.0	V	1-3
L22	EXT ID B10	54					
L23	EXT ID B11	55					
L24	EXT ID B12	56					
L25	EXT ID B13	57					
L26	EXT ID B14	58					
L27	EXT ID B15	59	-18.0	-1.5	0.0	V	1-3
L28	EXT IN LD1	70	-18.0	-1.2	0.0	V	1-3
L29	EXT IN LD0	71					
L30	EXT OUT SEL1	72					
L31	EXT OUT SEL0	73	-18.0	-1.2	0.0	V	1-3
L32	EXT ID B0	74	-18.0	-1.5	0.0	V	1-3
L33	EXT ID B1	76					
L34	EXT ID B2	77					
L35	EXT ID B3	78					
L36	EXT ID B4	79					
L37	EXT ID B5	80					
L38	EXT ID B6	82					
L39	EXT ID B7	83					
L40	DEV ID 5	84	-18.0	-1.5	0.0	V	1-3

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TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
L41	DEV ID 4	85	-18.0	-1.5	0.0	V	1-3
L42	DEV ID 3	86					
L43	DEV ID 2	87					
L44	DEV ID 1	88					
L45	DEV ID 0	89					
L46	IIU	91	-18.0	-1.5	0.0	V	1-3

NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	Off	NA	L1-L46
Power supply current limit	NA	NA	L1-L46
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1-L46
Pattern voltage high, (VH) low, (VL)	NA NA	NA NA	L1-L46 L1-L46

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input voltage measurement	NA	± 75 mV	L1-L11
	NA	± 60 mV	L12,L13
	NA	± 75 mV	L14
	NA	± 60 mV	L15-L17
	NA	± 75 mV	L18
	NA	± 60 mV	L19
	NA	± 75 mV	L20-L27
	NA	± 60 mV	L28-L31
	NA	± 75 mV	L32-L46

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XII and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
M1	PSFC	1	11.0	20.6	uF	1-4
M2	PSFC	51	6.6	12.4	uF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M1,M2
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M1,M2
Capacitance measurement	NA NA	± 480 nF ± 290 nF	M1 M2

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XIII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

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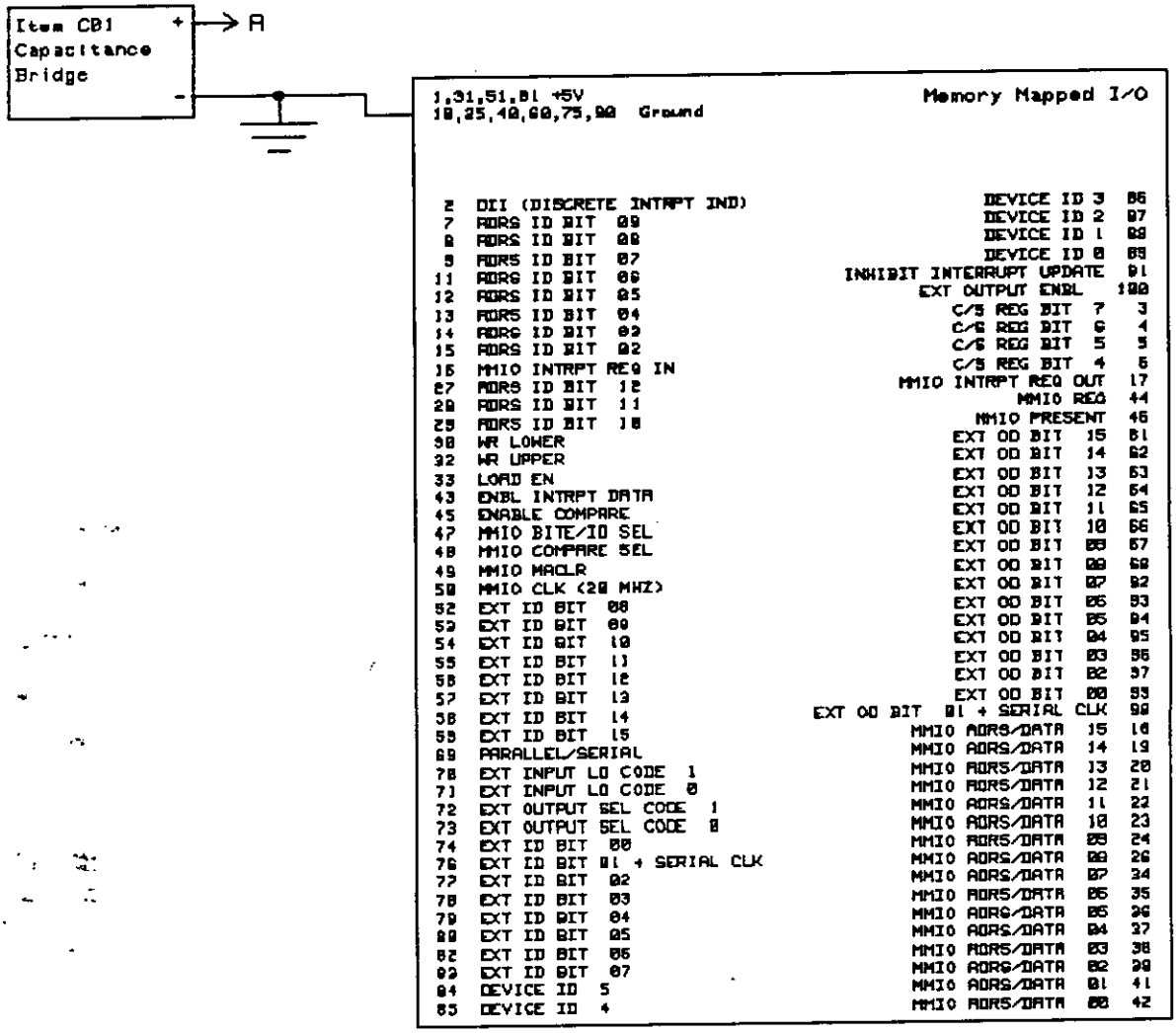


FIGURE 10. Power supply filter capacitance test circuit for table XIII.

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TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	Icc	1,31, 51,81	750	1800	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	2.7 A	NA	N
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	N N
Power supply current measurement	NA	± 52.5 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate (400 nS)	1-3
P2	Functional test maximum power supply voltage maximum pattern rate (400 nS)	1-3

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	2.7 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 20 mV	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

3. Abbreviations used are defined in 6.1.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (400 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (400 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (400 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (400 nS) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1,Q3 Q2,Q4
Power supply current limit	2.7 A	NA	Q1-Q4
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 Q1-Q4 Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 100 mV	Q1,Q3
high, (Voh)	NA	± 100 mV	Q2,Q4
low, (Vol)	NA	± 100 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

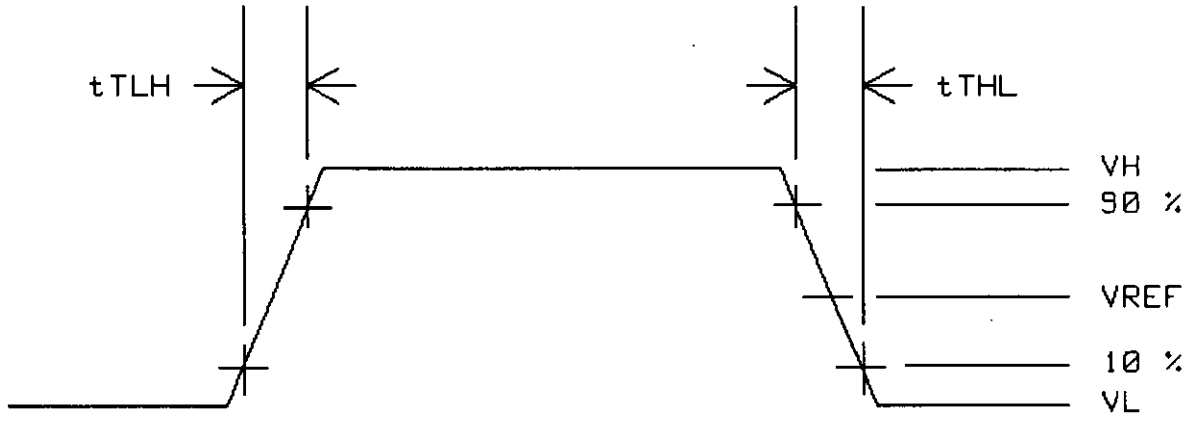
(b) The output compare levels shall be as follows:

High level: $Vref \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq Vref$

(c) The dynamic parameters are found in table XVII.

3. The load circuit for this test is specified on figure 9.

4. I/O waveforms are as referenced on figure 11. All of the input waveforms shall have the following characteristics:



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NOTES: - Continued.

Where:

(a) VH and VL are as specified in the test tolerance table.

(b) $t_{TLH} = t_{THL} \leq 10$ nS.

(c) Vref = 1.6 V.

5. This test shall be performed as indicated in table XVI.

6. Abbreviations used are defined in 6.1.

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TABLE XVII. Dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	MMIO CLK to MMIO REQ	-	-	-	37	37	37	nS	11
2	tpd	MMIO CLK to MMIO REQ	-	-	-	39	39	39	nS	11
3	tpd	MMIO CLK to C/S RG B5	-	-	-	66	66	66	nS	11
4	tpd	EXT OUT EN to C/S RG B5	-	-	-	55	55	55	nS	11
5	tpd	MMIO CLK to EXT OD B0	-	-	-	73	73	73	nS	11
5	tpd	MMIO CLK to EXT OD B1	-	-	-	73	73	73	nS	11
5	tpd	MMIO CLK to EXT OD B2	-	-	-	73	73	73	nS	11
5	tpd	MMIO CLK to EXT OD B3	-	-	-	73	73	73	nS	11
5	tpd	MMIO CLK to EXT OD B4	-	-	-	73	73	73	nS	11
5	tpd	MMIO CLK to EXT OD B5	-	-	-	73	73	73	nS	11
5	tpd	MMIO CLK to EXT OD B6	-	-	-	73	73	73	nS	11

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
5	tpd	MMIO CLK to EXT OD B7	-	-	-	73	73	73	nS	11
5	tpd	MMIO CLK to EXT OD B8	-	-	-	73	73	73	nS	11
5	tpd	MMIO CLK to EXT OD B9	-	-	-	73	73	73	nS	11
5	tpd	MMIO CLK to EXT OD B10	-	-	-	73	73	73	nS	11
5	tpd	MMIO CLK to EXT OD B11	-	-	-	73	73	73	nS	11
5	tpd	MMIO CLK to EXT OD B12	-	-	-	73	73	73	nS	11
5	tpd	MMIO CLK to EXT OD B13	-	-	-	73	73	73	nS	11
5	tpd	MMIO CLK to EXT OD B14	-	-	-	73	73	73	nS	11
5	tpd	MMIO CLK to EXT OD B15	-	-	-	73	73	73	nS	11
6	tpd	EXT OUT EN to EXT OD B0	-	-	-	30	30	30	nS	11
6	tpd	EXT OUT EN to EXT OD B1	-	-	-	30	30	30	nS	11
6	tpd	EXT OUT EN to EXT OD B2	-	-	-	30	30	30	nS	11

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
6	tpd	EXT OUT EN to EXT OD B3	-	-	-	30	30	30	nS	11
6	tpd	EXT OUT EN to EXT OD B4	-	-	-	30	30	30	nS	11
6	tpd	EXT OUT EN to EXT OD B5	-	-	-	30	30	30	nS	11
6	tpd	EXT OUT EN to EXT OD B6	-	-	-	30	30	30	nS	11
6	tpd	EXT OUT EN to EXT OD B7	-	-	-	30	30	30	nS	11
6	tpd	EXT OUT EN to EXT OD B8	-	-	-	30	30	30	nS	11
6	tpd	EXT OUT EN to EXT OD B9	-	-	-	30	30	30	nS	11
6	tpd	EXT OUT EN to EXT OD B10	-	-	-	30	30	30	nS	11
6	tpd	EXT OUT EN to EXT OD B11	-	-	-	30	30	30	nS	11
6	tpd	EXT OUT EN to EXT OD B12	-	-	-	30	30	30	nS	11
6	tpd	EXT OUT EN to EXT OD B13	-	-	-	30	30	30	nS	11
6	tpd	EXT OUT EN to EXT OD B14	-	-	-	30	30	30	nS	11

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
6	tpd	EXT OUT EN to EXT OD B15	-	-	-	30	30	30	nS	11
7 1/	tpd	MMIO CLK to EXT OD B8	-	-	-	73	73	73	nS	11
7	tpd	MMIO CLK to EXT OD B9	-	-	-	73	73	73	nS	11
7	tpd	MMIO CLK to EXT OD B10	-	-	-	73	73	73	nS	11
7	tpd	MMIO CLK to EXT OD B11	-	-	-	73	73	73	nS	11
7	tpd	MMIO CLK to EXT OD B12	-	-	-	73	73	73	nS	11
7	tpd	MMIO CLK to EXT OD B13	-	-	-	73	73	73	nS	11
7	tpd	MMIO CLK to EXT OD B14	-	-	-	73	73	73	nS	11
7	tpd	MMIO CLK to EXT OD B15	-	-	-	73	73	73	nS	11
8 1/	tpd	EXT OUT EN to EXT OD B8	-	-	-	30	30	30	nS	11
8	tpd	EXT OUT EN to EXT OD B9	-	-	-	30	30	30	nS	11
8	tpd	EXT OUT EN to EXT OD B10	-	-	-	30	30	30	nS	11

1/ For these tests pins 41, 42, 72, and 73 shall be set to 1.

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
8 1/	tpd	EXT OUT EN to EXT OD B11	-	-	-	30	30	30	nS	11
8	tpd	EXT OUT EN to EXT OD B12	-	-	-	30	30	30	nS	11
8	tpd	EXT OUT EN to EXT OD B13	-	-	-	30	30	30	nS	11
8	tpd	EXT OUT EN to EXT OD B14	-	-	-	30	30	30	nS	11
8	tpd	EXT OUT EN to EXT OD B15	-	-	-	30	30	30	nS	11
9	tpd	EXT OD B1 to EXT OD B0	-	-	-	53	53	53	nS	11
10	tpd	LOAD EN to C/S RG B7	-	-	-	33	33	33	nS	11
11	tpd	MMIO CLK to C/S RG B7	-	-	-	75	75	75	nS	11
12	tpd	LOAD EN to C/S RG B6	-	-	-	33	33	33	nS	11
13	tpd	MMIO CLK to C/S RG B6	-	-	-	75	75	75	nS	11
14	tpd	MMIO CLK to RD DAT 0	-	-	-	73	73	73	nS	11

1/ For these tests pins 41, 42, 72, and 73 shall be set to 1.

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
14	tpd	MMIO CLK to RD DAT 1	-	-	-	73	73	73	nS	11
14	tpd	MMIO CLK to RD DAT 2	-	-	-	73	73	73	nS	11
14	tpd	MMIO CLK to RD DAT 3	-	-	-	73	73	73	nS	11
14	tpd	MMIO CLK to RD DAT 4	-	-	-	73	73	73	nS	11
14	tpd	MMIO CLK to RD DAT 5	-	-	-	73	73	73	nS	11
14	tpd	MMIO CLK to RD DAT 6	-	-	-	73	73	73	nS	11
14	tpd	MMIO CLK to RD DAT 7	-	-	-	73	73	73	nS	11
14	tpd	MMIO CLK to RD DAT 8	-	-	-	73	73	73	nS	11
14	tpd	MMIO CLK to RD DAT 9	-	-	-	73	73	73	nS	11
14	tpd	MMIO CLK to RD DAT 10	-	-	-	73	73	73	nS	11
14	tpd	MMIO CLK to RD DAT 11	-	-	-	73	73	73	nS	11
14	tpd	MMIO CLK to RD DAT 12	-	-	-	73	73	73	nS	11

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
14	tpd	MMIO CLK to RD DAT 13	-	-	-	73	73	73	nS	11
14	tpd	MMIO CLK to RD DAT 14	-	-	-	73	73	73	nS	11
14	tpd	MMIO CLK to RD DAT 15	-	-	-	73	73	73	nS	11
15	tpd	MMIO CLK to RD DAT 0	-	-	-	73	73	73	nS	11
15	tpd	MMIO CLK to RD DAT 1	-	-	-	73	73	73	nS	11
15	tpd	MMIO CLK to RD DAT 2	-	-	-	73	73	73	nS	11
15	tpd	MMIO CLK to RD DAT 3	-	-	-	73	73	73	nS	11

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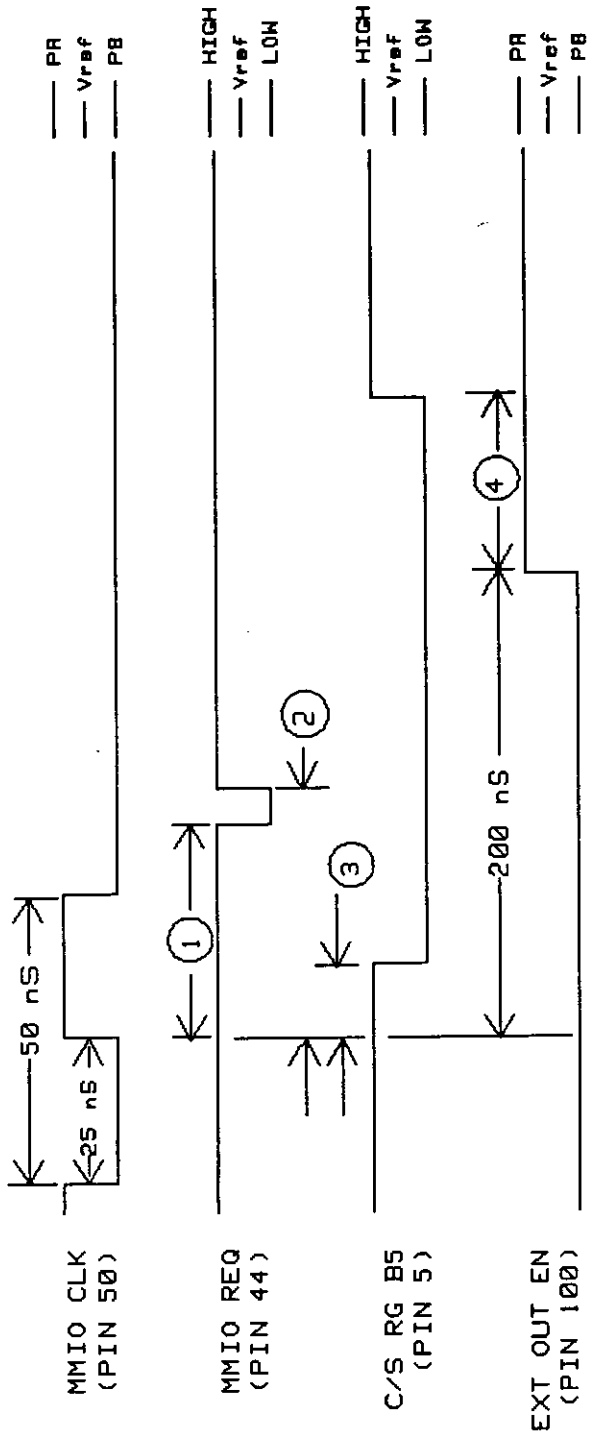


FIGURE 11a. Dynamic waveforms for table XVII.

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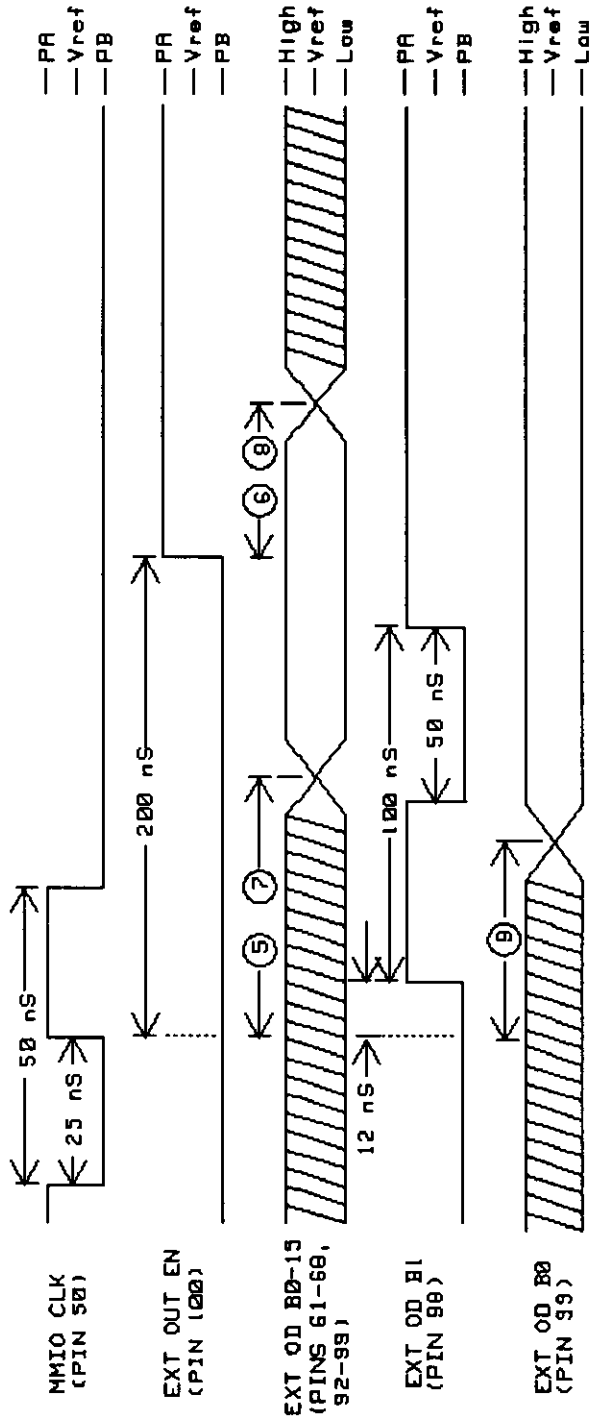


FIGURE 11b. Dynamic waveforms for table XVII.

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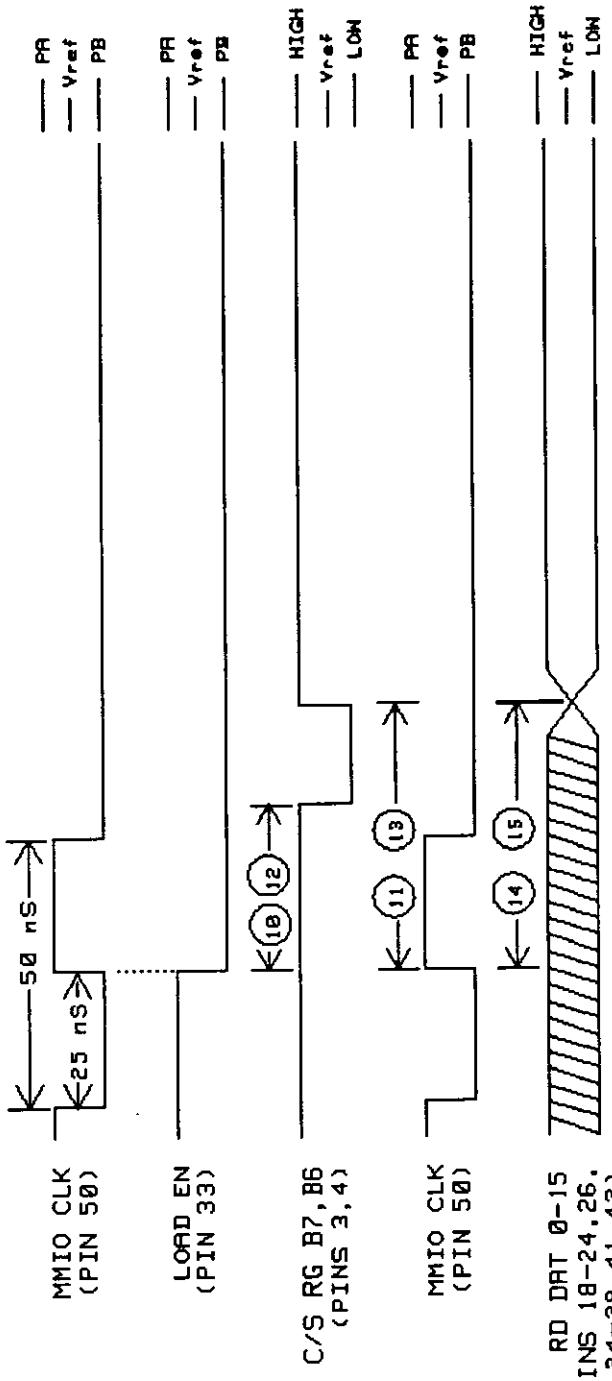


FIGURE 11c. Dynamic waveforms for table XVII.

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TABLE XVIII. Input and output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
Inputs						
R1	DII	2	4	100	pF	1-4
R2	AD ID B09	7	2	75	pF	1-4
R3	AD ID B08	8				
R4	AD ID B07	9				
R5	AD ID B06	11				
R6	AD ID B05	12				
R7	AD ID B04	13				
R8	AD ID B03	14				
R9	AD ID B02	15	2	75	pF	1-4
R10	MMIO IRQ IN	16	6	100	pF	1-4
R11	AD ID B12	27	2	75	pF	1-4
R12	AD ID B11	28	2	50	pF	1-4
R13	AD ID B10	29	2	75	pF	1-4
R14	WR LOWER	30	6	100	pF	1-4
R15	WR UPPER	32	6	100	pF	1-4
R16	LOAD EN	33	4	75	pF	1-4
R17	EN INT DAT	43	4	125	pF	1-4
R18	EN CMP	45	6	100	pF	1-4
R19	MMIO B/ID SE	47	2	75	pF	1-4
R20	MMIO CMP SEL	48	2	75	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R21	MMIO MACLR	49	2	100	pF	1-4
R22	MMIO CLK	50	4	75	pF	1-4
R23	EXT ID B8	52	2	75	pF	1-4
R24	EXT ID B9	53				
R25	EXT ID B10	54				
R26	EXT ID B11	55				
R27	EXT ID B12	56				
R28	EXT ID B13	57				
R29	EXT ID B14	58				
R30	EXT ID B15	59	2	75	pF	1-4
R31	PAR/SER	69	4	200	pF	1-4
R32	EXT IN LD1	70	2	75	pF	1-4
R33	EXT IN LD0	71				
R34	EXT OUT SEL1	72				
R35	EXT OUT SEL0	73	2	75	pF	1-4
R36	EXT ID B0	74	4	75	pF	1-4
R37	EXT ID B1	76	6	75	pF	1-4
R38	EXT ID B2	77	2	75	pF	1-4
R39	EXT ID B3	78	2	50	pF	1-4
R40	EXT ID B4	79	2	75	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R41	EXT ID B5	80	2	75	pF	1-4
R42	EXT ID B6	82				
R43	EXT ID B7	83				
R44	DEV ID 5	84				
R45	DEV ID 4	85				
R46	DEV ID 3	86				
R47	DEV ID 2	87				
R48	DEV ID 1	88				
R49	DEV ID 0	89	2	75	pF	1-4
R50	IIU	91	2	100	pF	1-4
R51	EXT OUT EN	100	6	175	pF	1-4
Bidirectionals						
R52	RD DAT 15	18	4	75	pF	1-4
R53	RD DAT 14	19				
R54	RD DAT 13	20	4	75	pF	1-4
R55	RD DAT 12	21	6	75	pF	1-4
R56	RD DAT 11	22				
R57	RD DAT 10	23	6	75	pF	1-4
R58	RD DAT 9	24	6	100	pF	1-4
R59	RD DAT 8	26				
R60	RD DAT 7	34	6	100	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R61	RD DAT 6	35	6	100	pF	1-4
R62	RD DAT 5	36				
R63	RD DAT 4	37	6	100	pF	1-4
R64	RD DAT 3	38	6	125	pF	1-4
R65	RD DAT 2	39	6	100	pF	1-4
R66	RD DAT 1	41	6	125	pF	1-4
R67	RD DAT 0	42	6	125	pF	1-4

NOTES:

1. Input and output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R67
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R67
Capacitance measurement	NA	± 4.80 pF	R1
	NA	± 3.65 pF	R2-R9
	NA	± 4.70 pF	R10
	NA	± 3.65 pF	R11
	NA	± 2.40 pF	R12
	NA	± 3.65 pF	R13
	NA	± 4.70 pF	R14,R15

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Capacitance measurement	NA	± 3.55 pF	R16
	NA	± 6.05 pF	R17
	NA	± 4.70 pF	R18
	NA	± 3.65 pF	R19,R20
	NA	± 4.90 pF	R21
	NA	± 3.55 pF	R22
	NA	± 3.65 pF	R23-R30
	NA	± 9.80 pF	R31
	NA	± 3.65 pF	R32-R35
	NA	± 3.55 pF	R36
	NA	± 3.45 pF	R37
	NA	± 3.65 pF	R38
	NA	± 2.40 pF	R39
	NA	± 3.65 pF	R40-R49
	NA	± 4.90 pF	R50
	NA	± 8.45 pF	R51
	NA	± 3.55 pF	R52-R54
	NA	± 3.45 pF	R55-R57
	NA	± 4.70 pF	R58-R63
	NA	± 5.95 pF	R64
	NA	± 4.70 pF	R65
	NA	± 5.95 pF	R66,R67

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. The test circuit shall be as indicated on figure 12.

4. Abbreviations used are defined in 6.1.

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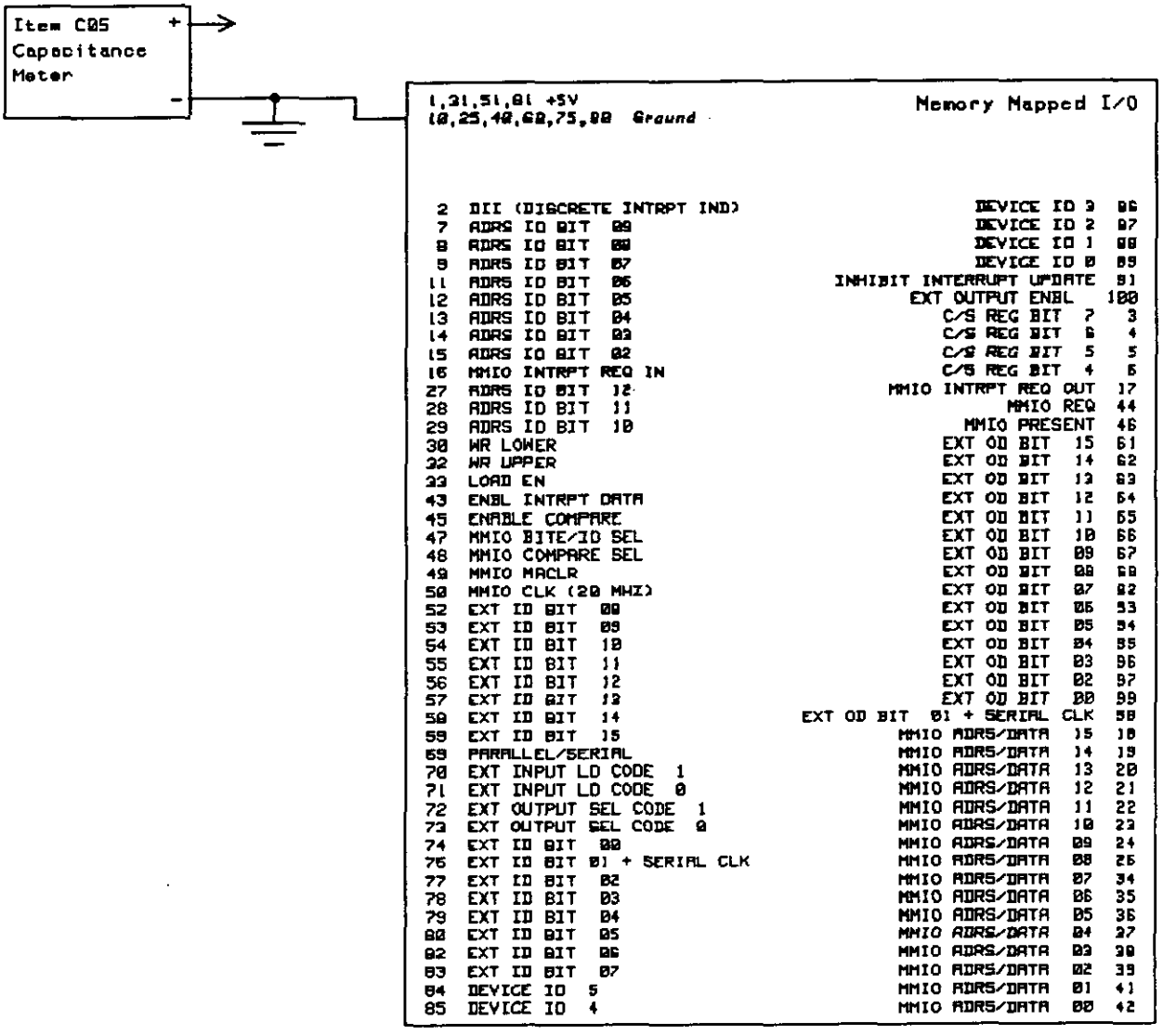


FIGURE 12. Input and output capacitance test circuit for table XVIII.

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TABLE XIX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
T1	PSTA	1	0	400	mV	1-4
T2	PSTA	51	0	400	mV	1-4

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	T1,T2
Power supply current limit	2.7 A	NA	T1,T2
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T1,T2 T1,T2
Power supply transient amplitude measurement	NA	± 20 mV	T1,T2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XIX.

(c) Vcc pins 1 and 31 shall be shorted together, pins 51 and 81 shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XIX.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 13 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 14.

4. Abbreviations used are defined in 6.1.

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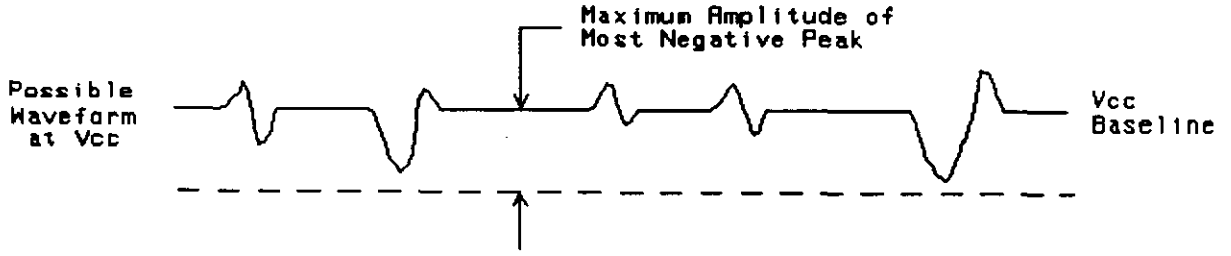
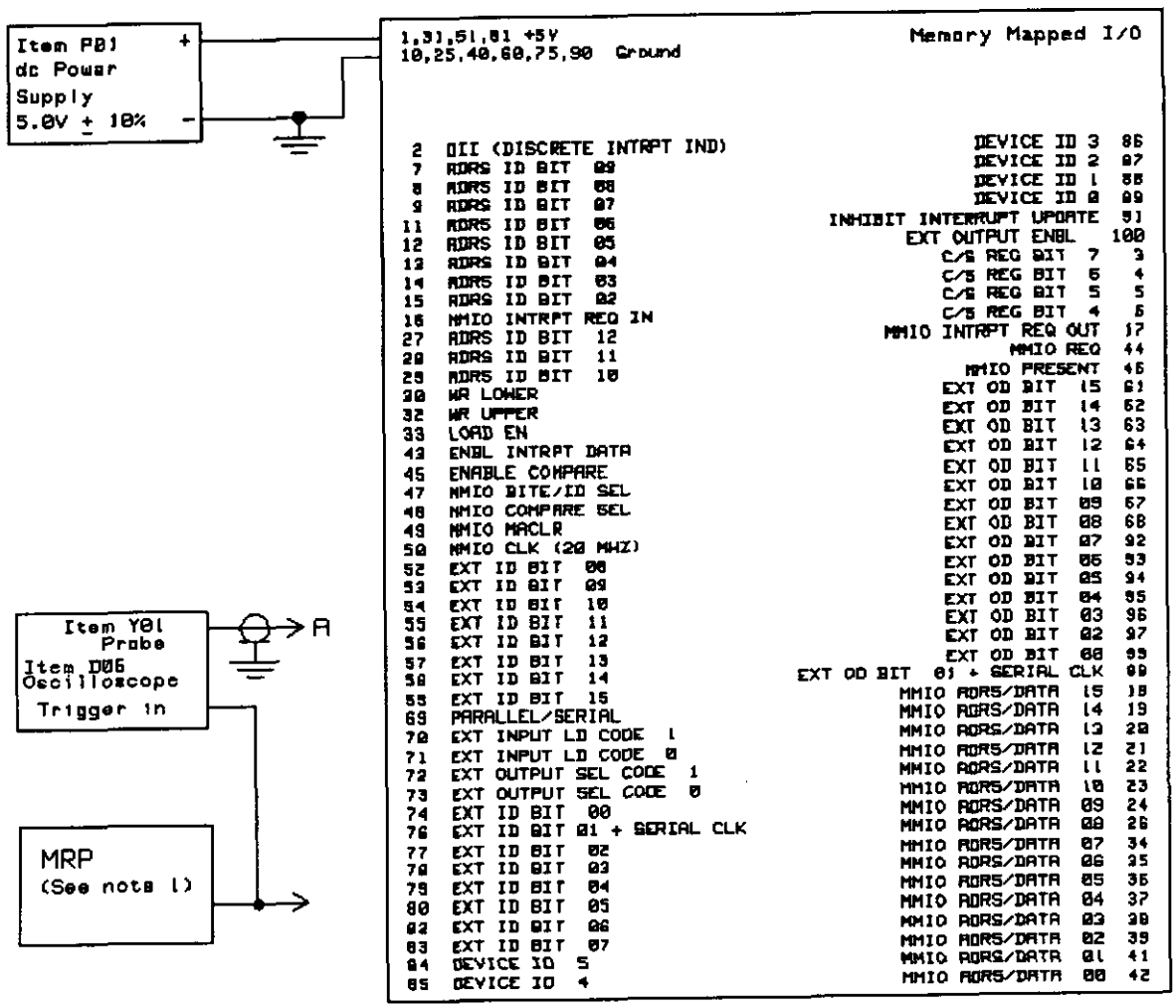


FIGURE 13. Power supply transient amplitude waveform for table XIX.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 14. Power supply transient amplitude test circuit for table XIX.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-28787/512	Modules, Standard Electronic, Memory Mapped I/O Module, Key Code BWG

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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